Smart City Ontology:
A Formal Representation of Interconnected Urban Subsystems

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Abstract

Urbanisation continues to accelerate globally, placing an increasing amount of pressure on cities to manage resources sustainably while improving the quality of life for their citizens. Smart Cities utilize digital technologies, the Internet of Things (IoT), and data-driven decision-making to address these challenges through an interconnected infrastructure. This study presents a Smart City Ontology designed to conceptualise critical information and subsystems across four primary domains: Smart Transportation, Smart Energy Grid, Smart Water Management, and Smart Healthcare. By providing a formalisation of enabling technologies and their complex relationships, this ontology serves as a shared vocabulary for developers building smart city knowledge graphs and automated decision support systems. The ontology was implemented in the Web Ontology Language (OWL) using Protégé. Validation through SPARQL queries demonstrates its ability to answer all 32 competency questions defined at the outset of the project.

Keywords: Smart City, Ontology, Internet of Things, Urban Infrastructure, Description Logic, OWL, SPARQL, Smart Transportation, Smart Energy Grid, Smart Water Management, Smart Healthcare.

Contents

1 Introduction

A Smart City is defined as an urban environment that employs information and communication (ICT), the Internet of Things (IoT), and data analytics to optimise the management of services and infrastructure. This includes transportation, energy, water management [2] [3]. As urban environments expand, an ontology serves as a vital knowledge resource to conceptualise the interconnected subsystems of modern infrastructure. An ontology formalises a shared conceptualisation of a domain through a structured vocabulary of classes, relations, instances, and axioms, enabling automated reasoning and knowledge exchange between diverse systems.

This research formalises the domains of Transportation, Energy, Water, and Healthcare, underpinned by IoT sensors and AI-driven decision support [4] [5]. Standardising this representation benefits urban planners and technology developers by providing a foundation for interoperable smart city frameworks. Urban planners, municipal governments, technology companies, policymakers, and researchers in this field would particularly benefit from such a structured knowledge base.

Smart City ontologies would benefit cities implementing smarter infrastructures, serving as a shared vocabulary for developers building smart city knowledge graphs and decision support systems [6],

The scope of this ontology is limited to identifying the four primary domains of a Smart City, defining the technologies and components that enable each domain, describing the relationships between systems within and across these domains, and outlining how each domain contributes to urban quality of life and sustainability.

The remainder of this report is structured as follows. Section 2 provides a literature review. Section 3 describes the methodology, data collection, and formalisation approach. Section 4 presents the full vocabulary and axioms. Section 5 presents the implementation in Protégé, SPARQL query results, and validation against competency questions. Section 6 concludes the report.

2 Literature Review

The foundational concept of Smart Cities presented in [1] identifies the economy, mobility, environment, people, living, and governance as the core dimensions of a Smart City. This is the foundation of our selection of Smart Transportation, Smart Energy Grid, Smart Water Management, and Smart Healthcare as the four main domains in our ontology. One of the earliest Smart City frameworks was introduced in [2], where urban intelligence was characterised across six functional dimensions. This characterisation shaped the scope of our competency questions, particularly those concerning transportation, energy, water, and healthcare, and reinforced our decision to model the Smart City as a composite of specific domain subsystems rather than one monolithic concept.

The IoT architecture described in [3] was key to structuring the technological infrastructure underlying the ontology. Our axioms are directly informed by their three-tier model of sensing, networking, and cloud analytics, which governs the IoT Platform, Edge Computing Node, Cloud Platform, and Urban Sensor Network classes, as well as the *transmitsData* and *processesDataFrom* properties that link domain-specific components to shared infrastructure. The work in [4] distinguishes between infrastructure-heavy domains such as transport and energy, and service-oriented ones such as healthcare. This distinction helped shape the depth and structure of the four chosen domains and the decision to represent cross-domain relationships through shared properties such as *integrates* and *communicatesWith*.

The axioms that link Traffic Management Systems, Connected Vehicles, and the IoT Platform were informed by [5], which discusses intelligent transportation systems and their role in emergency response. The treatment of adaptive signal control and variable message signs provided the basis for defining those classes and their relationships. The communication layers, metering infrastructure, and distributed energy resources that form the backbone of the Smart Energy Grid domain are discussed in [6], and the vocabulary and axioms for this domain were drawn from that work. The integration of renewable energy resources and demand response systems is further elaborated in [7], which informed the Virtual Power Plant and Microgrid classes.

The Smart Water Management framework presented in [8] provided the basis for the vocabulary and axioms in that domain, covering SCADA-based monitoring, smart metering, and distribution network management. This includes the relationship between the Water Treatment Plant, Pump Station, Leak Detection

System, and Water Distribution Network. The work in [9] further extends this by discussing data-driven water network optimisation, informing the Asset Management System class and its maintenance relationships.

A comprehensive taxonomy of smart health technologies in urban areas, including wearable sensors, telemedicine platforms, hospital information systems, and remote patient monitoring systems, is presented in [10]. This shaped the hierarchy in the Smart Healthcare domain and its axioms. The application of AI and machine learning in clinical decision-making, forming the basis of the AI Diagnostic Tool instance and the Clinical Decision Support class, is surveyed in [11].

The detailed look into IoT deployments across urban environments in [12] was instrumental in developing the axioms for the 5G and Urban Sensor Network classes, and the relationships between Cloud Computing and Edge Computing Nodes. The discussion of smart city ontology frameworks in [13] guided the structure of our class hierarchy and the use of properties to express cross-domain integration.

A critical perspective on data-driven urbanism is provided in [14], highlighting the governance and privacy implications of pervasive urban sensing. The axioms relating the Privacy Framework and Cybersecurity System to the IoT Platform are grounded in principles which that work articulates, acknowledging that data governance is a first-class concern in Smart City design. Security challenges particularly in smart city IoT deployments are further addressed in [15], supporting the Cybersecurity System class and the *securedBy* and *secures* properties.

The role of Digital Twins in smart urban infrastructure management is examined in [16], motivating the Digital Twin class and its properties. Edge computing architectures for latency-sensitive smart city applications are explained in [17], emphasising the distinction between Edge Computing Node and Cloud Platform classes. The interoperability challenges between heterogeneous smart city systems, which encourage the use of ontologies as integration frameworks, are addressed in [18]. The standardisation of smart city data models and ontologies discussed in [19] provide normative grounds for the vocabulary choices made in this work. Finally, the sociotechnical dimensions of smart city governance, including citizen participation and municipal accountability, are explored in [20], explaining the Citizen and Municipality classes, as well as their service-access relationships.

3 Materials and Methods

3.1 Methodology

A structure inspired by Ontology Development 101 principles [21] was followed that served as an objective approach for development. This ensures that the data extracted is based on factual statements discovered in the literature. Data was collected by iterative research of existing IoT and urban frameworks. The objective approach means that all axioms, classes, and properties were derived from domain facts rather than personal assertions.

To validate the scope, a set of 32 competency questions (CQs) was developed. These questions span across the four domains and the shared IoT infrastructure, covering queries from basic definitional questions (e.g., “What is a Smart City?”) to more specific operational ones (e.g., “How does leak detection work in a Smart Water network?”).

The ontology is formalised in the Web Ontology Language (OWL) using Protégé 5.5.0, with axioms translated into First-Order Logic (FOL) and Description Logic (DL) for logical consistency. The HermiT 1.4.3 reasoner was then used to verify satisfiability and infer class memberships.

3.2 Data Collection

Our data collection approach consisted of multiple iterations of researching relevant literature and domain sources. Sources consulted include academic journal articles on smart city IoT architectures, smart grid communications, smart water management, and smart healthcare systems. Existing smart city ontology work [13] and international standard frameworks [19] were also reviewed to inform vocabulary choices. Data was extracted as factual statements from these sources and directly translated into ontology axioms, instances, and properties.

The 32 competency questions defined at the outset guided which concepts and relationships were necessary to represent. These questions are listed fully in Section 4.1.

3.3 Formalisation of Ontology

The ontology language is OWL 2 with RDF/XML serialisation. This choice allows for a greater expressive vocabulary, better-defined structural constraints, and more efficient reuse of existing ontologies compared to RDFS only. All axioms were first stated in natural language (as presented in the list of axioms), then expressed formally in both First-Order Logic (FOL) and Description Logic (DL). This dual representation provides logical precision while remaining accessible and easy to interpret.

Protégé 5.5.0 was used as the ontology editing environment. The OWLViz plugin provided visual representations of the class hierarchy, and HermiT 1.4.3 served as the OWL reasoner. The implementation was carried out on a standard development laptop running the Protégé desktop application.

4 Modelling of Ontology

4.1 Competency Questions

1. What is a Smart City?
2. What are the primary domains of a Smart City?
3. What technologies enable a Smart City?
4. What is Smart Transportation?
5. What are the components of a Smart Transportation system?
6. How does traffic management work in a Smart City?
7. What types of smart mobility solutions exist?
8. How is public transit integrated in a Smart City?
9. How are electric vehicles supported in a Smart City?
10. What is a Smart Energy Grid?
11. What are the components of a Smart Grid?
12. How is energy consumption monitored?
13. How is renewable energy integrated?
14. How does demand response work?
15. What is Smart Water Management?
16. What are the components of a Smart Water system?
17. How is water quality monitored?
18. How does leak detection work?
19. How is wastewater treated and recycled?
20. What is Smart Healthcare?
21. What are the components of a Smart Healthcare system?
22. How are health records managed?
23. What wearable devices are used?
24. How does telemedicine function?

25. How are ambulances managed?
26. What is the IoT Platform's role?
27. What types of sensors are used?
28. What is a Digital Twin?
29. How is data collected and processed?
30. What are the privacy and security concerns?
31. What are the benefits of a Smart City?
32. What are the challenges of implementing a Smart City?

4.2 Vocabulary of the Domain

The vocabulary consists of classes and relations representing the core domains.

- **Classes:** Smart City, Urban Environment, IoT Platform, Edge Computing Node, Cloud Platform, Data Analytics Engine, 5G Network, SCADA System, Cybersecurity System, Privacy Framework, Municipality, Citizen, Urban Sensor Network, Digital Twin, Smart City Infrastructure, Smart Transportation, Traffic Management System, Traffic Sensor, Video Detection System, Traffic Signal Controller, Variable Message Sign, Real-Time Data Stream, Connected Vehicle, Public Transit System, Smart Ticketing, Passenger Information System, Fleet Management System, Parking Management System, Micro-Mobility, Traffic Flow, Smart Energy Grid, Advanced Metering Infrastructure, Smart Meter, Energy Management System, Substation, Distribution Automation System, Distributed Energy Resource, Smart Inverter, Battery Storage System, Demand Response System, Virtual Power Plant, Load Balancing System, Energy Consumption, Smart Water Management, Water Treatment Plant, Water Quality Sensor, Water Distribution Network, Smart Water Meter, Pump Station, Water Pipe Network, Leak Detection System, Reservoir, Asset Management System, Wastewater Treatment Plant, Stormwater Management System, Water Consumption, Smart Healthcare, Electronic Health Record, Hospital Information System, Telemedicine Platform, Wearable Health Sensor, Remote Patient Monitoring System, Ambulance Dispatch System, Smart Pharmacy, Health Analytics Platform, Clinical Decision Support System, Population Health, Patient Data, Data.
- **Relations:** componentOf, typeOf, uses, monitors, communicatesWith, transmitsData, generates, domainOf, controls, communicatesVia, usesDataFrom, manages, usedBy, integrates, analysesDataFrom, virtualRepresentationOf, storesEnergyFrom, storesDataFrom, secures, securedBy, produces, processesDataFrom, poweredBy, optimises, operates, measuresEnergyProducedBy, managedBy, maintains, governsDataHandlingIn, providedBy, enablesCommunicationAcross, convertsEnergyGeneratedBy, controlledBy, alerts, accessesServices, triggers, storesDataIn, integratesDataFrom, isAnInstanceOf, stores, from.

4.3 List of Axioms

The following formalizations represent the core logic of the smart city infrastructure.

4.3.1 First-Order Logic (FOL) Representation

1. Smart City is a type of Urban Environment
   $\forall x(SmartCity(x) \to UrbanEnvironment(x))$
2. Smart Transportation is a domain of Smart City
   $\forall x(SmartTransportation(x) \to \exists y(SmartCity(y) \land isADomainOf(x, y)))$
3. Smart Energy Grid is a domain of Smart City
   $\forall x(SmartEnergyGrid(x) \to \exists y(SmartCity(y) \land isADomainOf(x, y)))$

4. Smart Water Management is a domain of Smart City
   $\forall x(SmartWaterManagement(x) \rightarrow \exists y(SmartCity(y) \wedge isADomainOf(x, y)))$

5. Smart Healthcare is a domain of Smart City
   $\forall x(SmartHealthcare(x) \rightarrow \exists y(SmartCity(y) \wedge isADomainOf(x, y)))$

6. IoT Platform integrates data from Urban Sensor Network
   $\forall x(IotPlatform(x) \rightarrow \exists y, z(UrbanSensorNetwork(y) \wedge integrates(x, z) \wedge Data(z) \wedge from(z, y)))$

7. Edge Computing Node processes data from IoT Platform
   $\forall x(EdgeComputingNode(x) \rightarrow \exists y, z(IotPlatform(y) \wedge processes(x, z) \wedge Data(z) \wedge from(z, y)))$

8. Cloud Platform stores data from IoT Platform
   $\forall x(CloudPlatform(x) \rightarrow \exists y, z(IotPlatform(y) \wedge stores(x, z) \wedge Data(z) \wedge from(z, y)))$

9. Data Analytics Engine analyses data from Cloud Platform
   $\forall x(DataAnalyticsEngine(x) \rightarrow \exists y, z(CloudPlatform(y) \wedge analyses(x, z) \wedge Data(z) \wedge from(z, y)))$

10. Digital Twin is a virtual representation of Smart City Infrastructure
    $\forall x(DigitalTwin(x) \rightarrow \exists y(SmartCityInfrastructure(y) \wedge isAVirtualRepresentationOf(x, y)))$

11. Cybersecurity System secures IoT Platform
    $\forall x(CybersecuritySystem(x) \rightarrow \exists y(IotPlatform(y) \wedge secures(x, y)))$

12. 5G Network enables communication across Smart City domains
    $\forall x(5GNetwork(x) \rightarrow \exists y(SmartCity(y) \wedge enablesCommunicationAcross(x, y)))$

13. Privacy Framework governs data handling in Smart City
    $\forall x(PrivacyFramework(x) \rightarrow \exists y(SmartCity(y) \wedge governsDataHandlingIn(x, y)))$

14. Municipality operates Smart City
    $\forall x(Municipality(x) \rightarrow \exists y(SmartCity(y) \wedge operates(x, y)))$

15. Citizen accesses services of Smart City
    $\forall x(Citizen(x) \rightarrow \exists y(SmartCity(y) \wedge accessesServicesOf(x, y)))$

16. Smart City is provided by Municipality
    $\forall x(SmartCity(x) \rightarrow \exists y(Municipality(y) \wedge isProvidedBy(x, y)))$

17. High Population Density Area is an instance of Urban Environment
    $UrbanEnvironment(highPopulationDensityArea)$

18. Residential Neighbourhood is an instance of Urban Environment
    $UrbanEnvironment(residentialNeighbourhood)$

19. Industrial Zone is an instance of Urban Environment
    $UrbanEnvironment(industrialZone)$

20. Urban Environment is managed by Municipality
    $\forall x(UrbanEnvironment(x) \rightarrow \exists y(Municipality(y) \wedge isManagedBy(x, y)))$

21. Urban Sensor Network is a component of Smart City
    $\forall x(UrbanSensorNetwork(x) \rightarrow \exists y(SmartCity(y) \wedge isAComponentOf(x, y)))$

22. Traffic Management System is a component of Smart Transportation
    $\forall x(TrafficManagementSystem(x) \rightarrow \exists y(SmartTransportation(y) \wedge isAComponentOf(x, y)))$

23. Traffic Sensor is a component of Traffic Management System
    $\forall x(TrafficSensor(x) \rightarrow \exists y(TrafficManagementSystem(y) \wedge isAComponentOf(x, y)))$

24. Inductive Loop Detector is an instance of Traffic Sensor
    $TrafficSensor(inductiveLoopDetector)$

25. Infrared Sensor is an instance of Traffic Sensor
$TrafficSensor(infraredSensor)$

26. Radar Sensor is an instance of Traffic Sensor
$TrafficSensor(radarSensor)$

27. Video Detection System is a type of Traffic Sensor
$\forall x(VideoDetectionSystem(x) \to TrafficSensor(x))$

28. CCTV Camera is an instance of Video Detection System
$VideoDetectionSystem(cctvCamera)$

29. Traffic Signal Controller uses data from Traffic Sensor
$\forall x(TrafficSignalController(x) \to \exists y(TrafficSensor(y) \land usesDataFrom(x, y)))$

30. Adaptive Signal Control is an instance of Traffic Signal Controller
$TrafficSignalController(adaptiveSignalControl)$

31. Adaptive Signal Control optimises Traffic Flow
optimises(adaptiveSignalControl,trafficFlow)

32. Variable Message Sign is a component of Traffic Management System
$\forall x(VariableMessageSign(x) \to \exists y(TrafficManagementSystem(y) \land isAComponentOf(x, y)))$

33. Traffic Management System generates Real-Time Data Stream
$\forall x(TrafficManagementSystem(x) \to \exists y(Real - timeDataStream(y) \land generates(x, y)))$

34. Connected Vehicle communicates with Traffic Management System
$\forall x(ConnectedVehicle(x) \to \exists y(TrafficManagementSystem(y) \land communicatesWith(x, y)))$

35. Autonomous Vehicle is an instance of Connected Vehicle
$ConnectedVehicle(autonomousVehicle)$

36. Electric Vehicle is an instance of Connected Vehicle
$ConnectedVehicle(electricVehicle)$

37. Vehicle-to-Infrastructure Communication is an instance of Connected Vehicle
$ConnectedVehicle(vehicle - to - infrastructureCommunication)$

38. Vehicle-to-Vehicle Communication is an instance of Connected Vehicle
$ConnectedVehicle(vehicle - to - vehicleCommunication)$

39. Public Transit System is a component of Smart Transportation
$\forall x(PublicTransitSystem(x) \to \exists y(SmartTransportation(y) \land isAComponentOf(x, y)))$

40. Bus Rapid Transit is an instance of Public Transit System
$PublicTransitSystem(busRapidTransit)$

41. Metro Rail is an instance of Public Transit System
$PublicTransitSystem(metroRail)$

42. Light Rail Transit is an instance of Public Transit System
$PublicTransitSystem(lightRailTransit)$

43. Smart Ticketing is a component of Public Transit System
$\forall x(SmartTicketing(x) \to \exists y(PublicTransitSystem(y) \land isAComponentOf(x, y)))$

44. Contactless Payment Terminal is an instance of Smart Ticketing
$SmartTicketing(contactlessPaymentTerminal)$

45. Passenger Information System is a component of Public Transit System
$\forall x(PassengerInformationSystem(x) \to \exists y(PublicTransitSystem(y) \land isAComponentOf(x, y)))$

46. Fleet Management System manages Public Transit System
$\forall x(FleetManagementSystem(x) \rightarrow \exists y(PublicTransitSystem(y) \wedge manages(x,y)))$

47. Fleet Management System communicates with Passenger Information System
$\forall x(FleetManagementSystem(x) \rightarrow \exists y(PassengerInformationSystem(y) \wedge communicatesWith(x,y)))$

48. Parking Management System is a component of Smart Transportation
$\forall x(ParkingManagementSystem(x) \rightarrow \exists y(SmartTransportation(y) \wedge isAComponentOf(x,y)))$

49. Smart Parking Sensor is an instance of Parking Management System
$ParkingManagementSystem(smartParkingSensor)$

50. Congestion Pricing System is an instance of Traffic Management System
$TrafficManagementSystem(congestionPricingSystem)$

51. Navigation Application is an instance of Traffic Management System
$TrafficManagementSystem(navigationApplication)$

52. Electric Vehicle uses Electric Vehicle Charging Station
uses(electricVehicle,electricVehicleChargingStation)

53. Ride-Sharing Platform is an instance of Micro-Mobility
$Micro-mobility(ride-sharingPlatform)$

54. Bicycle Sharing System is an instance of Micro-Mobility
$Micro-mobility(bicycleSharingSystem)$

55. Micro-Mobility is a component of Smart Transportation
$\forall x(Micro-mobility(x) \rightarrow \exists y(SmartTransportation(y) \wedge isAComponentOf(x,y)))$

56. Smart Transportation transmits data via IoT Platform
$\forall x(SmartTransportation(x) \rightarrow \exists y(IotPlatform(y) \wedge transmitsDataVia(x,y)))$

57. Traffic Management System A monitors Peak Traffic Hours
monitors(trafficManagementSystemA,peakTrafficHours)

58. Traffic Management System B monitors Off-Peak Hours
monitors(trafficManagementSystemB,off-peakHours)

59. Advanced Metering Infrastructure is a component of Smart Energy Grid
$\forall x(AdvancedMeteringInfrastructure(x) \rightarrow \exists y(SmartEnergyGrid(y) \wedge isAComponentOf(x,y)))$

60. Smart Meter is a component of Advanced Metering Infrastructure
$\forall x(SmartMeter(x) \rightarrow \exists y(AdvancedMeteringInfrastructure(y) \wedge isAComponentOf(x,y)))$

61. Smart Meter monitors Energy Consumption
$\forall x(SmartMeter(x) \rightarrow \exists y(EnergyConsumption(y) \wedge monitors(x,y)))$

62. Smart Meter A communicates via Power Line Communication
communicatesVia(smartMeterA,powerLineCommunication)

63. Energy Management System communicates with Smart Meter
$\forall x(EnergyManagementSystem(x) \rightarrow \exists y(SmartMeter(y) \wedge communicatesWith(x,y)))$

64. Substation is a component of Smart Energy Grid
$\forall x(Substation(x) \rightarrow \exists y(SmartEnergyGrid(y) \wedge isAComponentOf(x,y)))$

65. Automated Substation is an instance of Substation
$Substation(automatedSubstation)$

66. Transformer is an instance of Substation
$Substation(transformer)$

67. Distribution Automation System controls Substation
$\forall x(DistributionAutomationSystem(x) \rightarrow \exists y(Substation(y) \wedge controls(x, y)))$

68. Distributed Energy Resource is a component of Smart Energy Grid
$\forall x(DistributedEnergyResource(x) \rightarrow \exists y(SmartEnergyGrid(y) \wedge isAComponentOf(x, y)))$

69. Solar Photovoltaic Panel is an instance of Distributed Energy Resource
$DistributedEnergyResource(solarPhotovoltaicPanel)$

70. Wind Turbine is an instance of Distributed Energy Resource
$DistributedEnergyResource(windTurbine)$

71. Smart Inverter is a component of Distributed Energy Resource
$\forall x(SmartInverter(x) \rightarrow \exists y(DistributedEnergyResource(y) \wedge isAComponentOf(x, y)))$

72. Smart Inverter A converts energy generated by Solar Photovoltaic Panel
convertsEnergyGeneratedBy(smartInverterA,solarPhotovoltaicPanel)

73. Battery Storage System stores energy from Distributed Energy Resource
$\forall x(BatteryStorageSystem(x) \rightarrow \exists y(DistributedEnergyResource(y) \wedge storesEnergyFrom(x, y)))$

74. Lithium-Ion Battery is an instance of Battery Storage System
$BatteryStorageSystem(lithium - ionBattery)$

75. Demand Response System manages Smart Energy Grid
$\forall x(DemandResponseSystem(x) \rightarrow \exists y(SmartEnergyGrid(y) \wedge manages(x, y)))$

76. Energy Management System controls Demand Response System
$\forall x(EnergyManagementSystem(x) \rightarrow \exists y(DemandResponseSystem(y) \wedge controls(x, y)))$

77. Home Energy Management System is an instance of Energy Management System
$EnergyManagementSystem(homeEnergyManagementSystem)$

78. Smart Appliance is controlled by Home Energy Management System
isControlledBy(smartAppliance,homeEnergyManagementSystem)

79. Electric Vehicle Charging Station is an instance of Smart Energy Grid
$SmartEnergyGrid(electricVehicleChargingStation)$

80. Microgrid is an instance of Smart Energy Grid
$SmartEnergyGrid(microgrid)$

81. Microgrid is powered by Distributed Energy Resource
isPoweredBy(microgrid,distributedEnergyResourceA)

82. Virtual Power Plant integrates Distributed Energy Resource
$\forall x(VirtualPowerPlant(x) \rightarrow \exists y(DistributedEnergyResource(y) \wedge integrates(x, y)))$

83. Virtual Power Plant is managed by Energy Management System
$\forall x(VirtualPowerPlant(x) \rightarrow \exists y(EnergyManagementSystem(y) \wedge isManagedBy(x, y)))$

84. Load Balancing System is a component of Smart Energy Grid
$\forall x(LoadBalancingSystem(x) \rightarrow \exists y(SmartEnergyGrid(y) \wedge isAComponentOf(x, y)))$

85. Outage Management System is an instance of Smart Energy Grid
$SmartEnergyGrid(outageManagementSystem)$

86. Net Metering System measures energy produced by Solar Photovoltaic Panel
measuresEnergyProducedBy(netMeteringSystem,solarPhotovoltaicPanel)

87. Carbon Emission Monitor is an instance of Smart Energy Grid
$SmartEnergyGrid(carbonEmissionMonitor)$

88. Smart Energy Grid generates Real-Time Data Stream
$\forall x(SmartEnergyGrid(x) \rightarrow \exists y(Real-timeDataStream(y) \wedge generates(x,y)))$

89. Smart Energy Grid transmits data via IoT Platform
$\forall x(SmartEnergyGrid(x) \rightarrow \exists y(IotPlatform(y) \wedge transmitsDataVia(x,y)))$

90. Substation A produces Grid Electricity
produces(substationA,gridElectricity)

91. Smart Energy Grid A optimises Carbon-Neutral Target
optimises(smartEnergyGridA,carbon-neutralTarget)

92. Power Line Communication is an instance of IoT Platform
$IotPlatform(powerLineCommunication)$

93. Water Treatment Plant is a component of Smart Water Management
$\forall x(WaterTreatmentPlant(x) \rightarrow \exists y(SmartWaterManagement(y) \wedge isAComponentOf(x,y)))$

94. Desalination Plant is an instance of Water Treatment Plant
$WaterTreatmentPlant(desalinationPlant)$

95. Water Quality Sensor is a component of Water Treatment Plant
$\forall x(WaterQualitySensor(x) \rightarrow \exists y(WaterTreatmentPlant(y) \wedge isAComponentOf(x,y)))$

96. pH Sensor is an instance of Water Quality Sensor
$WaterQualitySensor(phSensor)$

97. Turbidity Sensor is an instance of Water Quality Sensor
$WaterQualitySensor(turbiditySensor)$

98. Chlorine Sensor is an instance of Water Quality Sensor
$WaterQualitySensor(chlorineSensor)$

99. SCADA System controls Water Treatment Plant
$\forall x(ScadaSystem(x) \rightarrow \exists y(WaterTreatmentPlant(y) \wedge controls(x,y)))$

100. SCADA System controls Pump Station
$\forall x(ScadaSystem(x) \rightarrow \exists y(PumpStation(y) \wedge controls(x,y)))$

101. SCADA System generates Real-Time Data Stream
$\forall x(ScadaSystem(x) \rightarrow \exists y(Real-timeDataStream(y) \wedge generates(x,y)))$

102. Water Distribution Network is a component of Smart Water Management
$\forall x(WaterDistributionNetwork(x) \rightarrow \exists y(SmartWaterManagement(y) \wedge isAComponentOf(x,y)))$

103. Smart Water Meter is a component of Water Distribution Network
$\forall x(SmartWaterMeter(x) \rightarrow \exists y(WaterDistributionNetwork(y) \wedge isAComponentOf(x,y)))$

104. Smart Water Meter monitors Water Consumption
$\forall x(SmartWaterMeter(x) \rightarrow \exists y(WaterConsumption(y) \wedge monitors(x,y)))$

105. Smart Water Meter communicates via IoT Platform
$\forall x(SmartWaterMeter(x) \rightarrow \exists y(IotPlatform(y) \wedge communicatesVia(x,y)))$

106. Pump Station is a component of Water Distribution Network
$\forall x(PumpStation(x) \rightarrow \exists y(WaterDistributionNetwork(y) \wedge isAComponentOf(x,y)))$

107. Water Pipe Network is a component of Water Distribution Network
$\forall x(WaterPipeNetwork(x) \rightarrow \exists y(WaterDistributionNetwork(y) \wedge isAComponentOf(x,y)))$

108. Leak Detection System monitors Water Distribution Network
$\forall x(LeakDetectionSystem(x) \rightarrow \exists y(WaterDistributionNetwork(y) \wedge monitors(x,y)))$

109. Pressure Sensor is an instance of Leak Detection System
$LeakDetectionSystem(pressureSensor)$

110. Flow Sensor is an instance of Leak Detection System
$LeakDetectionSystem(flowSensor)$

111. Reservoir is a component of Smart Water Management
$\forall x(Reservoir(x) \rightarrow \exists y(SmartWaterManagement(y) \wedge isAComponentOf(x, y)))$

112. Elevated Storage Tank is an instance of Reservoir
$Reservoir(elevatedStorageTank)$

113. Asset Management System maintains Water Distribution Network
$\forall x(AssetManagementSystem(x) \rightarrow \exists y(WaterDistributionNetwork(y) \wedge maintains(x, y)))$

114. Wastewater Treatment Plant is a component of Smart Water Management
$\forall x(WastewaterTreatmentPlant(x) \rightarrow \exists y(SmartWaterManagement(y) \wedge isAComponentOf(x, y)))$

115. Sewer Network is an instance of Wastewater Treatment Plant
$WastewaterTreatmentPlant(sewerNetwork)$

116. Water Recycling System is an instance of Wastewater Treatment Plant
$WastewaterTreatmentPlant(waterRecyclingSystem)$

117. Stormwater Management System is a component of Smart Water Management
$\forall x(StormwaterManagementSystem(x) \rightarrow \exists y(SmartWaterManagement(y) \wedge isAComponentOf(x, y)))$

118. Green Infrastructure is an instance of Stormwater Management System
$StormwaterManagementSystem(greenInfrastructure)$

119. Rain Sensor is an instance of Stormwater Management System
$StormwaterManagementSystem(rainSensor)$

120. Flood Detection Sensor is an instance of Stormwater Management System
$StormwaterManagementSystem(floodDetectionSensor)$

121. Emergency Alert System is an instance of Smart Water Management
$SmartWaterManagement(emergencyAlertSystem)$

122. Leak Detection System A alerts Emergency Alert System
alerts(leakDetectionSystemA,emergencyAlertSystem)

123. Smart Water Management transmits data via IoT Platform
$\forall x(SmartWaterManagement(x) \rightarrow \exists y(IotPlatform(y) \wedge transmitsDataVia(x, y)))$

124. Water Treatment Plant A produces Potable Water
produces(waterTreatmentPlantA,potableWater)

125. Wastewater Treatment Plant A produces Recycled Wastewater
produces(wastewaterTreatmentPlantA,recycledWastewater)

126. Emergency State triggers Emergency Alert System
triggers(emergencyState,emergencyAlertSystem)

127. SCADA System A monitors Normal Operating Condition
monitors(scadaSystemA,normalOperatingCondition)

128. Flood Detection Sensor monitors High-Risk Flood Zone
monitors(floodDetectionSensor,high-riskFloodZone)

129. Planned Maintenance Window is managed by Asset Management System A
isManagedBy(plannedMaintenanceWindow,assetManagementSystemA)

130. Electronic Health Record is a component of Smart Healthcare
$\forall x(ElectronicHealthRecord(x) \rightarrow \exists y(SmartHealthcare(y) \wedge isAComponentOf(x, y)))$

131. Hospital Information System manages Electronic Health Record
$\forall x(HospitalInformationSystem(x) \rightarrow \exists y(ElectronicHealthRecord(y) \wedge manages(x, y)))$

132. Telemedicine Platform uses Electronic Health Record
$\forall x(TelemedicinePlatform(x) \rightarrow \exists y(ElectronicHealthRecord(y) \wedge uses(x, y)))$

133. Telemedicine Platform communicates via 5G Network
$\forall x(TelemedicinePlatform(x) \rightarrow \exists y(5gNetwork(y) \wedge communicatesVia(x, y)))$

134. Wearable Health Sensor generates Patient Data
$\forall x(WearableHealthSensor(x) \rightarrow \exists y(PatientData(y) \wedge generates(x, y)))$

135. Heart Rate Monitor is an instance of Wearable Health Sensor
$WearableHealthSensor(heartRateMonitor)$

136. Blood Pressure Monitor is an instance of Wearable Health Sensor
$WearableHealthSensor(bloodPressureMonitor)$

137. Glucose Monitor is an instance of Wearable Health Sensor
$WearableHealthSensor(glucoseMonitor)$

138. Remote Patient Monitoring System uses data from Wearable Health Sensor
$\forall x(RemotePatientMonitoringSystem(x) \rightarrow \exists y(WearableHealthSensor(y) \wedge usesDataFrom(x, y)))$

139. Remote Patient Monitoring System communicates with Electronic Health Record
$\forall x(RemotePatientMonitoringSystem(x) \rightarrow \exists y(ElectronicHealthRecord(y) \wedge communicatesWith(x, y)))$

140. Ambulance Dispatch System is a component of Smart Healthcare
$\forall x(AmbulanceDispatchSystem(x) \rightarrow \exists y(SmartHealthcare(y) \wedge isAComponentOf(x, y)))$

141. Smart Ambulance is an instance of Ambulance Dispatch System
$AmbulanceDispatchSystem(smartAmbulance)$

142. Smart Ambulance communicates with Ambulance Dispatch System A
communicatesWith(smartAmbulance,ambulanceDispatchSystemA)

143. Smart Pharmacy is a component of Smart Healthcare
$\forall x(SmartPharmacy(x) \rightarrow \exists y(SmartHealthcare(y) \wedge isAComponentOf(x, y)))$

144. Automated Prescription Dispenser is an instance of Smart Pharmacy
$SmartPharmacy(automatedPrescriptionDispenser)$

145. Health Analytics Platform analyses data from Hospital Information System
$\forall x(HealthAnalyticsPlatform(x) \rightarrow \exists y(HospitalInformationSystem(y) \wedge analysesDataFrom(x, y)))$

146. Clinical Decision Support System uses Health Analytics Platform
$\forall x(ClinicalDecisionSupportSystem(x) \rightarrow \exists y(HealthAnalyticsPlatform(y) \wedge uses(x, y)))$

147. AI Diagnostic Tool is an instance of Clinical Decision Support System
$ClinicalDecisionSupportSystem(aiDiagnosticTool)$

148. Medical Imaging System is an instance of Hospital Information System
$HospitalInformationSystem(medicalImagingSystem)$

149. Biometric Identification System is an instance of Hospital Information System
$HospitalInformationSystem(biometricIdentificationSystem)$

150. Community Health Kiosk is an instance of Smart Healthcare
$SmartHealthcare(communityHealthKiosk)$

151. Community Health Kiosk uses Electronic Health Record A
uses(communityHealthKiosk,electronicHealthRecordA)

152. Elderly Care Monitoring System is an instance of Smart Healthcare
$SmartHealthcare(elderlyCareMonitoringSystem)$

153. Public Health Surveillance System is an instance of Smart Healthcare
$SmartHealthcare(publicHealthSurveillanceSystem)$

154. Public Health Surveillance System monitors Population Health
monitors(publicHealthSurveillanceSystem,populationHealth)

155. Vaccination Management System is an instance of Smart Healthcare
$SmartHealthcare(vaccinationManagementSystem)$

156. Smart Healthcare transmits data via IoT Platform
$\forall x(SmartHealthcare(x) \rightarrow \exists y(IotPlatform(y) \land transmitsDataVia(x, y)))$

157. Health Analytics Platform stores data in Cloud Platform
$\forall x(HealthAnalyticsPlatform(x) \rightarrow \exists y(CloudPlatform(y) \land storesDataIn(x, y)))$

158. Wearable Health Sensor communicates via 5G Network
$\forall x(WearableHealthSensor(x) \rightarrow \exists y(5gNetwork(y) \land communicatesVia(x, y)))$

159. Remote Patient Monitoring System is secured by Cybersecurity System
$\forall x(RemotePatientMonitoringSystem(x) \rightarrow \exists y(CybersecuritySystem(y) \land isSecuredBy(x, y)))$

160. Traffic Flow is an instance of Smart Transportation
$SmartTransportation(trafficFlow)$

161. Traffic Management System A is an instance of Traffic Management System
$TrafficManagementSystem(trafficManagementSystemA)$

162. Traffic Management System B is an instance of Traffic Management System
$TrafficManagementSystem(trafficManagementSystemB)$

163. Peak Traffic Hours is an instance of Smart Transportation
$SmartTransportation(peakTrafficHours)$

164. Off-Traffic Hours is an instance of Smart Transportation
$SmartTransportation(off - TrafficHours)$

165. Smart Meter A is an instance of Smart Meter
$SmartMeter(smartMeterA)$

166. smartInverterA is an instance of Smart Inverter
$SmartInverter(smartInverterA)$

167. Distributed Energy Resource A is an instance of Distributed Energy Resource
$DistributedEnergyResource(distributedEnergyResourceA)$

168. Net Metering System is an instance of Smart Energy Grid
$SmartEnergyGrid(netMeteringSystem)$

169. Substation A is an instance of Substation
$Substation(substationA)$

170. Grid Electricty is an instance of Smart Energy Grid
$SmartEnergyGrid(gridElectricity)$

171. Smart Energy Grid A is an instance of Smart Energy Grid
$SmartEnergyGrid(smartEnergyGridA)$

172. Carbon-neutral Target is an instance of Energy Management System
$EnergyManagementSystem(carbonNeutralTarget)$

173. Leak Detection System A is an instance of Leak Detection System
$LeakDetectionSystem(leakDetectionSystemA)$

174. Water Treatment Plant A is an instance of Water Treatment Plant
$WaterTreatmentPlant(waterTreatmentPlantA)$

175. Potable Water is an instance of Water Pipe Network
$WaterPipeNetwork(potableWater)$

176. Recycled Wastewater is an instance of Water Pipe Network
$WaterPipeNetwork(recycledWastewater)$

177. Wastewater Treatment Plant A is an instance of Wastewater Treatment Plant
$WastewaterTreatmentPlant(wastewaterTreatmentPlantA)$

178. Emergency State is an instance of SmartCity
$SmartCity(emergencyState)$

179. Scada System A is an instance of Scada System
$ScadaSystem(scadaSystemA)$

180. Normal Operating Conditions is an instance of Smart City
$SmartCity(normalOperatingConditions)$

181. High-risk Flood Zone is an instance of Stormwater Management System
$StormwaterManagementSystem(high-riskFloodZone)$

182. Planned Maintenance is an instance of Smart City
$SmartCity(plannedMaintenance)$

183. Asset Management System A is an instance of B
$AssetManagementSystem(assetManagementSystemA)$

184. Ambulance Dispatch System A is an instance of Ambulance Dispatch System
$AmbulanceDispatchSystem(ambulanceDispatchSystemA)$

185. Electronic Health Record A is an instance of Electronic Health Record
$ElectronicHealthRecord(electronicHealthRecordA)$

186. Population Health is an instance of Smart Healthcare
$SmartHealthcare(populationHealth)$

4.3.2 Description Logic (DL) Representation

1. Smart City is a type of Urban Environment
$SmartCity \subseteq UrbanEnvironment$

2. Smart Transportation is a domain of Smart City
$SmartTransportation \subseteq \exists isADomainOf.SmartCity$

3. Smart Energy Grid is a domain of Smart City
$SmartEnergyGrid \subseteq \exists isADomainOf.SmartCity$

4. Smart Water Management is a domain of Smart City
$SmartWaterManagement \subseteq \exists isADomainOf.SmartCity$

5. Smart Healthcare is a domain of Smart City
$SmartHealthcare \subseteq \exists isADomainOf.SmartCity$

6. IoT Platform integrates data from Urban Sensor Network
   $IotPlatform \subseteq \exists integrates.Data \cap \exists from.UrbanSensorNetwork$
7. Edge Computing Node processes data from IoT Platform
   $EdgeComputingNode \subseteq \exists processes.Data \cap \exists from.IotPlatform$
8. Cloud Platform stores data from IoT Platform
   $CloudPlatform \subseteq \exists stores.Data \cap \exists from.IotPlatform$
9. Data Analytics Engine analyses data from Cloud Platform
   $DataAnalyticsEngine \subseteq \exists analyses.Data \cap \exists from.CloudPlatform$
10. Digital Twin is a virtual representation of Smart City Infrastructure
    $DigitalTwin \subseteq \exists isAVirtualRepresentationOf.SmartCityInfrastructure$
11. Cybersecurity System secures IoT Platform
    $CybersecuritySystem \subseteq \exists secures.IotPlatform$
12. 5G Network enables communication across Smart City domains
    $5GNetwork \subseteq \exists enablesCommunicationAcross.SmartCity$
13. Privacy Framework governs data handling in Smart City
    $PrivacyFramework \subseteq \exists governsDataHandlingIn.SmartCity$
14. Municipality operates Smart City
    $Municipality \subseteq \exists operates.SmartCity$
15. Citizen accesses services of Smart City
    $Citizen \subseteq \exists accessesServicesOf.SmartCity$
16. Smart City is provided by Municipality
    $SmartCity \subseteq \exists isProvidedBy.Municipality$
17. High Population Density Area is an instance of Urban Environment
    $UrbanEnvironment(highPopulationDensityArea)$
18. Residential Neighbourhood is an instance of Urban Environment
    $UrbanEnvironment(residentialNeighbourhood)$
19. Industrial Zone is an instance of Urban Environment
    $UrbanEnvironment(industrialZone)$
20. Urban Environment is managed by Municipality
    $UrbanEnvironment \subseteq \exists isManagedBy.Municipality$
21. Urban Sensor Network is a component of Smart City
    $UrbanSensorNetwork \subseteq \exists isAComponentOf.SmartCity$
22. Traffic Management System is a component of Smart Transportation
    $TrafficManagementSystem \subseteq \exists isAComponentOf.SmartTransportation$
23. Traffic Sensor is a component of Traffic Management System
    $TrafficSensor \subseteq \exists isAComponentOf.TrafficManagementSystem$
24. Inductive Loop Detector is an instance of Traffic Sensor
    $TrafficSensor(inductiveLoopDetector)$
25. Infrared Sensor is an instance of Traffic Sensor
    $TrafficSensor(infraredSensor)$
26. Radar Sensor is an instance of Traffic Sensor
    $TrafficSensor(radarSensor)$

27. Video Detection System is a type of Traffic Sensor
$VideoDetectionSystem \subseteq TrafficSensor$

28. CCTV Camera is an instance of Video Detection System
$VideoDetectionSystem(cctvCamera)$

29. Traffic Signal Controller uses data from Traffic Sensor
$TrafficSignalController \subseteq \exists usesDataFrom.TrafficSensor$

30. Adaptive Signal Control is an instance of Traffic Signal Controller
$TrafficSignalController(adaptiveSignalControl)$

31. Adaptive Signal Control optimises Traffic Flow
optimises(adaptiveSignalControl,trafficFlow)

32. Variable Message Sign is a component of Traffic Management System
$VariableMessageSign \subseteq \exists isAComponentOf.TrafficManagementSystem$

33. Traffic Management System generates Real-Time Data Stream
$TrafficManagementSystem \subseteq \exists generates.Real - timeDataStream$

34. Connected Vehicle communicates with Traffic Management System
$ConnectedVehicle \subseteq \exists communicatesWith.TrafficManagementSystem$

35. Autonomous Vehicle is an instance of Connected Vehicle
$ConnectedVehicle(autonomousVehicle)$

36. Electric Vehicle is an instance of Connected Vehicle
$ConnectedVehicle(electricVehicle)$

37. Vehicle-to-Infrastructure Communication is an instance of Connected Vehicle
$ConnectedVehicle(vehicle - to - infrastructureCommunication)$

38. Vehicle-to-Vehicle Communication is an instance of Connected Vehicle
$ConnectedVehicle(vehicle - to - vehicleCommunication)$

39. Public Transit System is a component of Smart Transportation
$PublicTransitSystem \subseteq \exists isAComponentOf.SmartTransportation$

40. Bus Rapid Transit is an instance of Public Transit System
$PublicTransitSystem(busRapidTransit)$

41. Metro Rail is an instance of Public Transit System
$PublicTransitSystem(metroRail)$

42. Light Rail Transit is an instance of Public Transit System
$PublicTransitSystem(lightRailTransit)$

43. Smart Ticketing is a component of Public Transit System
$SmartTicketing \subseteq \exists isAComponentOf.PublicTransitSystem$

44. Contactless Payment Terminal is an instance of Smart Ticketing
$SmartTicketing(contactlessPaymentTerminal)$

45. Passenger Information System is a component of Public Transit System
$PassengerInformationSystem \subseteq \exists isAComponentOf.PublicTransitSystem$

46. Fleet Management System manages Public Transit System
$FleetManagementSystem \subseteq \exists manages.PublicTransitSystem$

47. Fleet Management System communicates with Passenger Information System
$FleetManagementSystem \subseteq \exists communicatesWith.PassengerInformationSystem$

48. Parking Management System is a component of Smart Transportation
$ParkingManagementSystem \subseteq \exists isAComponentOf.SmartTransportation$
49. Smart Parking Sensor is an instance of Parking Management System
$ParkingManagementSystem(smartParkingSensor)$
50. Congestion Pricing System is an instance of Traffic Management System
$TrafficManagementSystem(congestionPricingSystem)$
51. Navigation Application is an instance of Traffic Management System
$TrafficManagementSystem(navigationApplication)$
52. Electric Vehicle uses Electric Vehicle Charging Station
uses(electricVehicle,electricVehicleChargingStation)
53. Ride-Sharing Platform is an instance of Micro-Mobility
$Micro-mobility(ride-sharingPlatform)$
54. Bicycle Sharing System is an instance of Micro-Mobility
$Micro-mobility(bicycleSharingSystem)$
55. Micro-Mobility is a component of Smart Transportation
$Micro-mobility \subseteq \exists isAComponentOf.SmartTransportation$
56. Smart Transportation transmits data via IoT Platform
$SmartTransportation \subseteq \exists transmitsDataVia.IotPlatform$
57. Traffic Management System A monitors Peak Traffic Hours
monitors(trafficManagementSystemA,peakTrafficHours)
58. Traffic Management System B monitors Off-Peak Hours
monitors(trafficManagementSystemB,off-peakHours)
59. Advanced Metering Infrastructure is a component of Smart Energy Grid
$AdvancedMeteringInfrastructure \subseteq \exists isAComponentOf.SmartEnergyGrid$
60. Smart Meter is a component of Advanced Metering Infrastructure
$SmartMeter \subseteq \exists isAComponentOf.AdvancedMeteringInfrastructure$
61. Smart Meter monitors Energy Consumption
$SmartMeter \subseteq \exists monitors.EnergyConsumption$
62. Smart Meter A communicates via Power Line Communication
communicatesVia(SmartMeter,powerLineCommunication)
63. Energy Management System communicates with Smart Meter
$EnergyManagementSystem \subseteq \exists communicatesWith.SmartMeter$
64. Substation is a component of Smart Energy Grid
$Substation \subseteq \exists isAComponentOf.SmartEnergyGrid$
65. Automated Substation is an instance of Substation
$Substation(automatedSubstation)$
66. Transformer is an instance of Substation
$Substation(transformer)$
67. Distribution Automation System controls Substation
$DistributionAutomationSystem \subseteq \exists controls.Substation$
68. Distributed Energy Resource is a component of Smart Energy Grid
$DistributedEnergyResource \subseteq \exists isAComponentOf.SmartEnergyGrid$

69. Solar Photovoltaic Panel is an instance of Distributed Energy Resource
$DistributedEnergyResource(solarPhotovoltaicPanel)$

70. Wind Turbine is an instance of Distributed Energy Resource
$DistributedEnergyResource(windTurbine)$

71. Smart Inverter is a component of Distributed Energy Resource
$SmartInverter \subseteq \exists isAComponentOf.DistributedEnergyResource$

72. Smart Inverter A converts energy generated by Solar Photovoltaic Panel
convertsEnergyGeneratedBy(smartInverterA,solarPhotovoltaicPanel)

73. Battery Storage System stores energy from Distributed Energy Resource
$BatteryStorageSystem \subseteq \exists storesEnergyFrom.DistributedEnergyResource$

74. Lithium-Ion Battery is an instance of Battery Storage System
$BatteryStorageSystem(lithium - ionBattery)$

75. Demand Response System manages Smart Energy Grid
$DemandResponseSystem \subseteq \exists manages.SmartEnergyGrid$

76. Energy Management System controls Demand Response System
$EnergyManagementSystem \subseteq \exists controls.DemandResponseSystem$

77. Home Energy Management System is an instance of Energy Management System
$EnergyManagementSystem(homeEnergyManagementSystem)$

78. Smart Appliance is controlled by Home Energy Management System
isControlledBy(smartAppliance,homeEnergyManagementSystem)

79. Electric Vehicle Charging Station is an instance of Smart Energy Grid
$SmartEnergyGrid(electricVehicleChargingStation)$

80. Microgrid is an instance of Smart Energy Grid
$SmartEnergyGrid(microgrid)$

81. Microgrid is powered by Distributed Energy Resource
isPoweredBy(microgrid,distributedEnergyResourceA)

82. Virtual Power Plant integrates Distributed Energy Resource
$VirtualPowerPlant \subseteq \exists integrates.DistributedEnergyResource$

83. Virtual Power Plant is managed by Energy Management System
$VirtualPowerPlant \subseteq \exists isManagedBy.EnergyManagementSystem$

84. Load Balancing System is a component of Smart Energy Grid
$LoadBalancingSystem \subseteq \exists isAComponentOf.SmartEnergyGrid$

85. Outage Management System is an instance of Smart Energy Grid
$SmartEnergyGrid(outageManagementSystem)$

86. Net Metering System measures energy produced by Solar Photovoltaic Panel
measuresEnergyProducedBy(netMeteringSystem,solarPhotovoltaicPanel)

87. Carbon Emission Monitor is an instance of Smart Energy Grid
$SmartEnergyGrid(carbonEmissionMonitor)$

88. Smart Energy Grid generates Real-Time Data Stream
$SmartEnergyGrid \subseteq \exists generates.Real - timeDataStream$

89. Smart Energy Grid transmits data via IoT Platform
$SmartEnergyGrid \subseteq \exists transmitsDataVia.IotPlatform$

90. Substation A produces Grid Electricity
produces(substationA,gridElectricity)

91. Smart Energy Grid A optimises Carbon-Neutral Target
optimises(smartEnergyGridA,carbon-neutralTarget)

92. Power Line Communication is an instance of IoT Platform
$IotPlatform(powerLineCommunication)$

93. Water Treatment Plant is a component of Smart Water Management
$WaterTreatmentPlant \subseteq \exists isAComponentOf.SmartWaterManagement$

94. Desalination Plant is an instance of Water Treatment Plant
$WaterTreatmentPlant(desalinationPlant)$

95. Water Quality Sensor is a component of Water Treatment Plant
$WaterQualitySensor \subseteq \exists isAComponentOf.WaterTreatmentPlant$

96. pH Sensor is an instance of Water Quality Sensor
$WaterQualitySensor(phSensor)$

97. Turbidity Sensor is an instance of Water Quality Sensor
$WaterQualitySensor(turbiditySensor)$

98. Chlorine Sensor is an instance of Water Quality Sensor
$WaterQualitySensor(chlorineSensor)$

99. SCADA System controls Water Treatment Plant
$ScadaSystem \subseteq \exists controls.WaterTreatmentPlant$

100. SCADA System controls Pump Station
$ScadaSystem \subseteq \exists controls.PumpStation$

101. SCADA System generates Real-Time Data Stream
$ScadaSystem \subseteq \exists generates.Real - timeDataStream$

102. Water Distribution Network is a component of Smart Water Management
$WaterDistributionNetwork \subseteq \exists isAComponentOf.SmartWaterManagement$

103. Smart Water Meter is a component of Water Distribution Network
$SmartWaterMeter \subseteq \exists isAComponentOf.WaterDistributionNetwork$

104. Smart Water Meter monitors Water Consumption
$SmartWaterMeter \subseteq \exists monitors.WaterConsumption$

105. Smart Water Meter communicates via IoT Platform
$SmartWaterMeter \subseteq \exists communicatesVia.IotPlatform$

106. Pump Station is a component of Water Distribution Network
$PumpStation \subseteq \exists isAComponentOf.WaterDistributionNetwork$

107. Water Pipe Network is a component of Water Distribution Network
$WaterPipeNetwork \subseteq \exists isAComponentOf.WaterDistributionNetwork$

108. Leak Detection System monitors Water Distribution Network
$LeakDetectionSystem \subseteq \exists monitors.WaterDistributionNetwork$

109. Pressure Sensor is an instance of Leak Detection System
$LeakDetectionSystem(pressureSensor)$

110. Flow Sensor is an instance of Leak Detection System
$LeakDetectionSystem(flowSensor)$

111. Reservoir is a component of Smart Water Management
$Reservoir \subseteq \exists isAComponentOf.SmartWaterManagement$
112. Elevated Storage Tank is an instance of Reservoir
$Reservoir(elevatedStorageTank)$
113. Asset Management System maintains Water Distribution Network
$AssetManagementSystem \subseteq \exists maintains.WaterDistributionNetwork$
114. Wastewater Treatment Plant is a component of Smart Water Management
$WastewaterTreatmentPlant \subseteq \exists isAComponentOf.SmartWaterManagement$
115. Sewer Network is an instance of Wastewater Treatment Plant
$WastewaterTreatmentPlant(sewerNetwork)$
116. Water Recycling System is an instance of Wastewater Treatment Plant
$WastewaterTreatmentPlant(waterRecyclingSystem)$
117. Stormwater Management System is a component of Smart Water Management
$StormwaterManagementSystem \subseteq \exists isAComponentOf.SmartWaterManagement$
118. Green Infrastructure is an instance of Stormwater Management System
$StormwaterManagementSystem(greenInfrastructure)$
119. Rain Sensor is an instance of Stormwater Management System
$StormwaterManagementSystem(rainSensor)$
120. Flood Detection Sensor is an instance of Stormwater Management System
$StormwaterManagementSystem(floodDetectionSensor)$
121. Emergency Alert System is an instance of Smart Water Management
$SmartWaterManagement(emergencyAlertSystem)$
122. Leak Detection System A alerts Emergency Alert System
alerts(leakDetectionSystemA,emergencyAlertSystem)
123. Smart Water Management transmits data via IoT Platform
$SmartWaterManagement \subseteq \exists transmitsDataVia.IotPlatform$
124. Water Treatment Plant A produces Potable Water
produces(waterTreatmentPlantA,potableWater)
125. Wastewater Treatment Plant A produces Recycled Wastewater
produces(wastewaterTreatmentPlantA,recycledWastewater)
126. Emergency State triggers Emergency Alert System
triggers(emergencyState,emergencyAlertSystem)
127. SCADA System A monitors Normal Operating Condition
monitors(scadaSystemA,normalOperatingCondition)
128. Flood Detection Sensor monitors High-Risk Flood Zone
monitors(floodDetectionSensor,high-riskFloodZone)
129. Planned Maintenance Window is managed by Asset Management System A
isManagedBy(plannedMaintenanceWindow,assetManagementSystemA)
130. Electronic Health Record is a component of Smart Healthcare
$ElectronicHealthRecord \subseteq \exists isAComponentOf.SmartHealthcare$
131. Hospital Information System manages Electronic Health Record
$HospitalInformationSystem \subseteq \exists manages.ElectronicHealthRecord$

132. Telemedicine Platform uses Electronic Health Record
$TelemedicinePlatform \subseteq \exists uses.ElectronicHealthRecord$

133. Telemedicine Platform communicates via 5G Network
$TelemedicinePlatform \subseteq \exists communicatesVia.5gNetwork$

134. Wearable Health Sensor generates Patient Data
$WearableHealthSensor \subseteq \exists generates.PatientData$

135. Heart Rate Monitor is an instance of Wearable Health Sensor
$WearableHealthSensor(heartRateMonitor)$

136. Blood Pressure Monitor is an instance of Wearable Health Sensor
$WearableHealthSensor(bloodPressureMonitor)$

137. Glucose Monitor is an instance of Wearable Health Sensor
$WearableHealthSensor(glucoseMonitor)$

138. Remote Patient Monitoring System uses data from Wearable Health Sensor
$RemotePatientMonitoringSystem \subseteq \exists usesDataFrom.WearableHealthSensor$

139. Remote Patient Monitoring System communicates with Electronic Health Record
$RemotePatientMonitoringSystem \subseteq \exists communicatesWith.ElectronicHealthRecord$

140. Ambulance Dispatch System is a component of Smart Healthcare
$AmbulanceDispatchSystem \subseteq \exists isAComponentOf.SmartHealthcare$

141. Smart Ambulance is an instance of Ambulance Dispatch System
$AmbulanceDispatchSystem(smartAmbulance)$

142. Smart Ambulance communicates with Ambulance Dispatch System A
communicatesWith(smartAmbulance,ambulanceDispatchSystemA)

143. Smart Pharmacy is a component of Smart Healthcare
$SmartPharmacy \subseteq \exists isAComponentOf.SmartHealthcare$

144. Automated Prescription Dispenser is an instance of Smart Pharmacy
$SmartPharmacy(automatedPrescriptionDispenser)$

145. Health Analytics Platform analyses data from Hospital Information System
$HealthAnalyticsPlatform \subseteq \exists analysesDataFrom.HospitalInformationSystem$

146. Clinical Decision Support System uses Health Analytics Platform
$ClinicalDecisionSupportSystem \subseteq \exists uses.HealthAnalyticsPlatform$

147. AI Diagnostic Tool is an instance of Clinical Decision Support System
$ClinicalDecisionSupportSystem(aiDiagnosticTool)$

148. Medical Imaging System is an instance of Hospital Information System
$HospitalInformationSystem(medicalImagingSystem)$

149. Biometric Identification System is an instance of Hospital Information System
$HospitalInformationSystem(biometricIdentificationSystem)$

150. Community Health Kiosk is an instance of Smart Healthcare
$SmartHealthcare(communityHealthKiosk)$

151. Community Health Kiosk uses Electronic Health Record A
uses(communityHealthKiosk,electronicHealthRecordA)

152. Elderly Care Monitoring System is an instance of Smart Healthcare
$SmartHealthcare(elderlyCareMonitoringSystem)$

153. Public Health Surveillance System is an instance of Smart Healthcare
$SmartHealthcare(publicHealthSurveillanceSystem)$

154. Public Health Surveillance System monitors Population Health
monitors(publicHealthSurveillanceSystem,populationHealth)

155. Vaccination Management System is an instance of Smart Healthcare
$SmartHealthcare(vaccinationManagementSystem)$

156. Smart Healthcare transmits data via IoT Platform
$SmartHealthcare \subseteq \exists transmitsDataVia.IotPlatform$

157. Health Analytics Platform stores data in Cloud Platform
$HealthAnalyticsPlatform \subseteq \exists storesDataIn.CloudPlatform$

158. Wearable Health Sensor communicates via 5G Network
$WearableHealthSensor \subseteq \exists communicatesVia.5gNetwork$

159. Remote Patient Monitoring System is secured by Cybersecurity System
$RemotePatientMonitoringSystem \subseteq \exists isSecuredBy.CybersecuritySystem$

160. Traffic Flow is an instance of Smart Transportation
$SmartTransportation(trafficFlow)$

161. Traffic Management System A is an instance of Traffic Management System
$TrafficManagementSystem(trafficManagementSystemA)$

162. Traffic Management System B is an instance of Traffic Management System
$TrafficManagementSystem(trafficManagementSystemB)$

163. Peak Traffic Hours is an instance of Smart Transportation
$SmartTransportation(peakTrafficHours)$

164. Off-Traffic Hours is an instance of Smart Transportation
$SmartTransportation(off - TrafficHours)$

165. Smart Meter A is an instance of Smart Meter
$SmartMeter(smartMeterA)$

166. smartInverterA is an instance of Smart Inverter
$SmartInverter(smartInverterA)$

167. Distributed Energy Resource A is an instance of Distributed Energy Resource
$DistributedEnergyResource(distributedEnergyResourceA)$

168. Net Metering System is an instance of Smart Energy Grid
$SmartEnergyGrid(netMeteringSystem)$

169. Substation A is an instance of Substation
$Substation(substationA)$

170. Grid Electricty is an instance of Smart Energy Grid
$SmartEnergyGrid(gridElectricity)$

171. Smart Energy Grid A is an instance of Smart Energy Grid
$SmartEnergyGrid(smartEnergyGridA)$

172. Carbon-neutral Target is an instance of Energy Management System
$EnergyManagementSystem(carbonNeutralTarget)$

173. Leak Detection System A is an instance of Leak Detection System
$LeakDetectionSystem(leakDetectionSystemA)$

174. Water Treatment Plant A is an instance of Water Treatment Plant
$WaterTreatmentPlant(waterTreatmentPlantA)$

175. Potable Water is an instance of Water Pipe Network
$WaterPipeNetwork(potableWater)$

176. Recycled Wastewater is an instance of Water Pipe Network
$WaterPipeNetwork(recycledWastewater)$

177. Wastewater Treatment Plant A is an instance of Wastewater Treatment Plant
$WastewaterTreatmentPlant(wastewaterTreatmentPlantA)$

178. Emergency State is an instance of SmartCity
$SmartCity(emergencyState)$

179. Scada System A is an instance of Scada System
$ScadaSystem(scadaSystemA)$

180. Normal Operating Conditions is an instance of Smart City
$SmartCity(normalOperatingConditions)$

181. High-risk Flood Zone is an instance of Stormwater Management System
$StormwaterManagementSystem(high-riskFloodZone)$

182. Planned Maintenance is an instance of Smart City
$SmartCity(plannedMaintenance)$

183. Asset Management System A is an instance of B
$AssetManagementSystem(assetManagementSystemA)$

184. Ambulance Dispatch System A is an instance of Ambulance Dispatch System
$AmbulanceDispatchSystem(ambulanceDispatchSystemA)$

185. Electronic Health Record A is an instance of Electronic Health Record
$ElectronicHealthRecord(electronicHealthRecordA)$

186. Population Health is an instance of Smart Healthcare
$SmartHealthcare(populationHealth)$

5 Experimental Results

5.1 Dataset

Table 1 summarises the primary data sources used to construct the ontology.

Table 1: Primary data sources used in ontology construction.

Source	Domain Coverage
Albino et al. [1]	Smart City definitions and dimensions
Giffinger et al. [2]	Smart City framework and six functional dimensions
Gubbi et al. [3]	IoT architecture: sensing, networking, cloud
Neirotti et al. [4]	Domain structure: infrastructure vs service domains
Djahel et al. [5]	Smart Transportation, ITS, traffic management
Fang et al. [6]	Smart Energy Grid, AMI, metering infrastructure
Beaudin & Zareipour [7]	Demand response, home energy management
Ramos & Monteiro [8]	Smart Water Management, SCADA monitoring
Savic & Vamvakeridou-Lyroudia [9]	Water network optimisation, asset management
Solanas et al. [10]	Smart Healthcare taxonomy, wearables, telemedicine
Dey et al. [11]	AI/ML in clinical decision support
Zanella et al. [12]	IoT for smart cities, sensor network axioms
Compton et al. [13]	SSN Ontology, semantic sensor web
Kitchin [14]	Privacy framework, data governance
Mosenia & Jha [15]	IoT security, cybersecurity systems
Tao et al. [16]	Digital Twin modelling
Shi et al. [17]	Edge computing architectures
Baldini et al. [18]	Interoperability in IoT systems
ITU-T [19]	Smart city data model standardisation
Nam & Pardo [20]	Sociotechnical dimensions, governance

5.2 Computer and Software Environments

The ontology was developed and implemented in OWL 2 with RDF/XML serialisation, using the open-source ontology editor Protégé 5.5.0. The OWLViz plugin was used to generate visual representations of the class hierarchy, supporting inspection of the four-domain structure and cross-domain relationships. The HermiT 1.4.3 OWL reasoner was applied to verify satisfiability, perform class classification, and infer class memberships across all 85 named individuals, confirming logical consistency of the full axiomatisation. SPARQL queries were executed using the SPARQL query interface built into Protégé to validate the ontology against the 32 competency questions. The software was used on a Lenovo Legion Y740 gaming laptop, with the following specifications: Intel i7-9750H CPU, Nvidia Geforce RTX 2070 Max-Q GPU, 16GB Ram, 2TB NVMe SSD, running Windows 11.

5.3 Results and Discussion

5.3.1 Ontology Structure in Protégé

Class Hierarchy

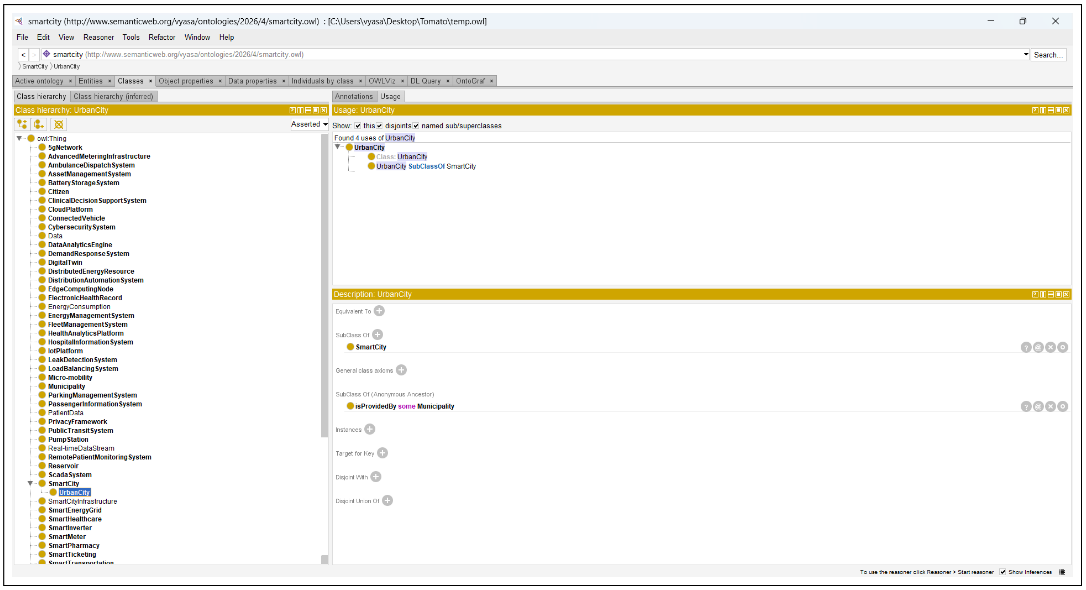


Figure 1: Top-level class hierarchy of the Smart City Ontology as displayed in the Protégé editor. The four domain classes are visible as direct subclasses branching from the core `SmartCity` class.

Figure 1 presents the top-level class hierarchy of the Smart City Ontology as visualised within the Protégé ontology editor. The hierarchy is rooted at `owl:Thing`, from which the primary domain classes (`SmartCity`, `SmartTransportation`, `SmartEnergyGrid`, `SmartWaterManagement`, `SmartHealthcare`) and the shared infrastructure classes (`IoTPlatform`, `EdgeComputingNode`, `CloudPlatform`, among others) are derived. In its entirety, the ontology comprises 68 classes organised into a hierarchy of depth 3.

Object Properties

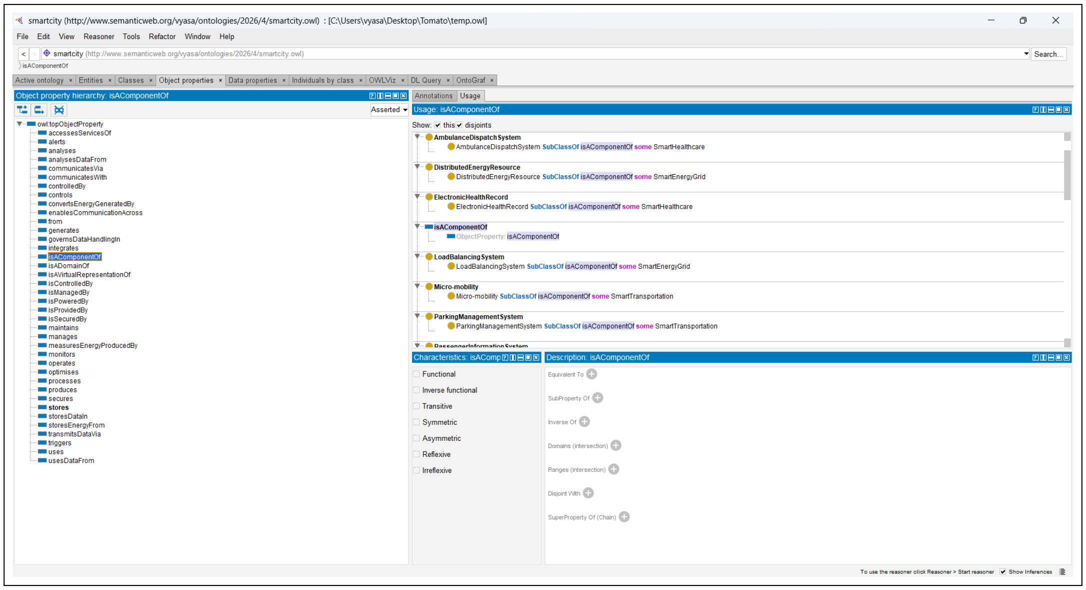


Figure 2: Object properties panel in Protégé showing the full list of 38 defined properties. The `isAComponentOf` property is selected, illustrating its domain and range annotations.

Figure 2 illustrates the object properties panel within Protégé. The `isAComponentOf` property, selected for examination, exemplifies the compositional relationships that associate subsystem classes with their respective parent domain classes. The ontology defines a total of 38 distinct object properties, collectively encoding the diverse inter-subsystem relationships characteristic of a smart city architecture.

Named Individuals

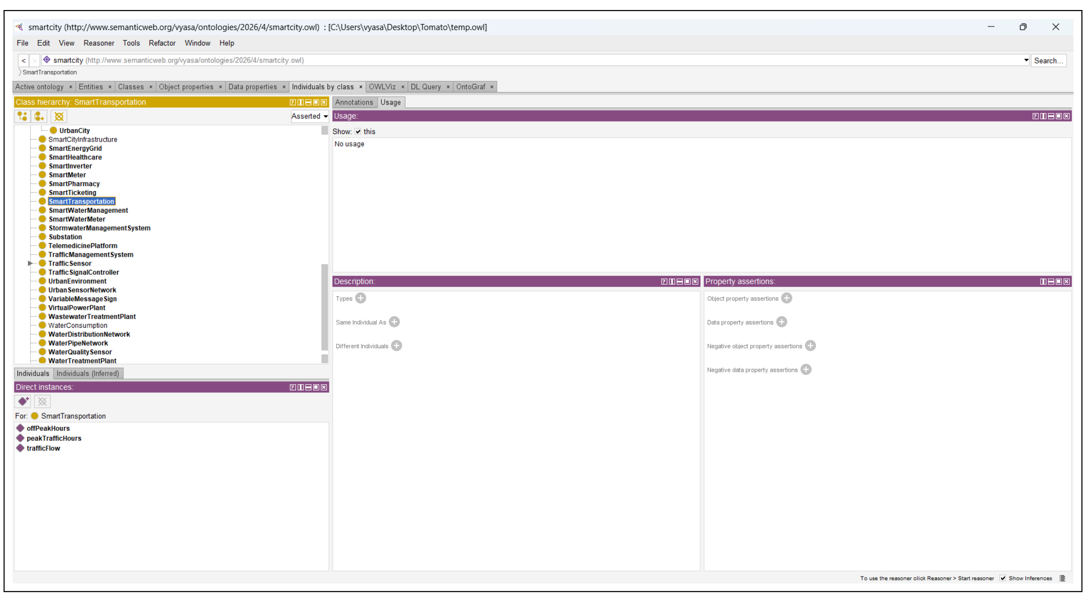


Figure 3: Named individuals panel in Protégé showing a subset of the 85 individuals defined in the ontology. Individuals from the Smart Transportation domain are highlighted, showing their inferred class memberships.

Figure 3 depicts a representative selection of named individuals as rendered in Protégé. Individuals belonging to the `SmartTransportation` domain, including `adaptiveSignalControl` and `electricVehicle`, are presented alongside their corresponding class memberships. Upon execution of the HermiT reasoner, all 85 individuals were successfully classified without the detection of any unsatisfiability errors, thereby affirming the internal coherence of the individual assertions.

Reasoner Output

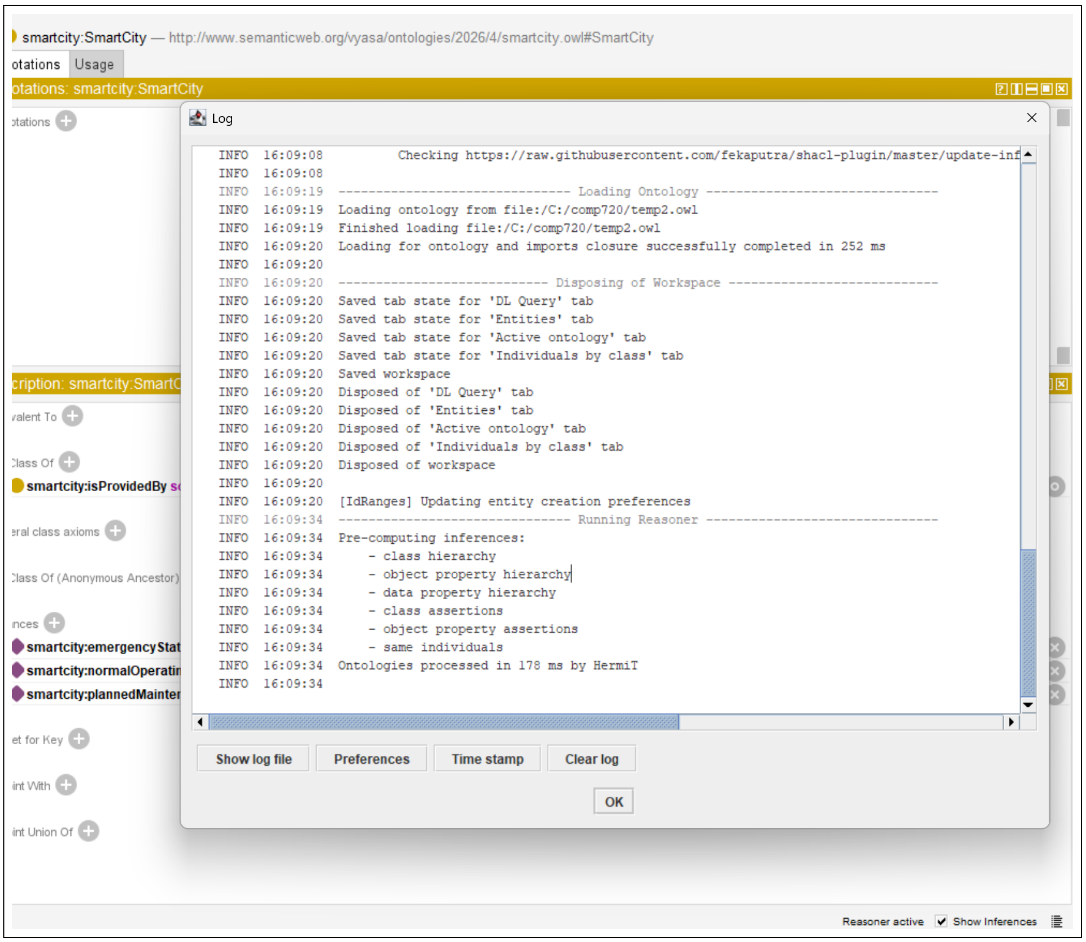


Figure 4: HermiT 1.4.3 reasoner output after running classification on the Smart City Ontology. No inconsistencies were detected, confirming the ontology is satisfiable.

Figure 4 presents the classification output produced by the HermiT reasoner following the completion of the reasoning process. The absence of unsatisfiable classes confirms that the ontology is logically consistent. Furthermore, the inferred class hierarchy is in full agreement with the asserted hierarchy, and all existential restrictions are correctly satisfied, demonstrating the semantic validity of the ontology's axiomatisation.

5.3.2 SPARQL Query Results

SPARQL queries were executed to validate the ontology against the 32 competency questions. Selected queries and their results are presented below. The prefix used throughout is:

```
PREFIX sc: <http://www.semanticweb.org/vyasa/ontologies/2026/4/smartcity.owl#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
```

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
```

Listing 1: Ontology prefix declaration

CQ1: What is a Smart City?

```
SELECT DISTINCT ?smartCityClass
WHERE {
  ?smartCityClass rdfs:subClassOf* sc:SmartCity .
}
```

Listing 2: SPARQL query for CQ1 – definition of a Smart City

Result: The query returns all classes that are subclasses of `sc:SmartCity`, identifying the class hierarchy rooted at the Smart City concept.

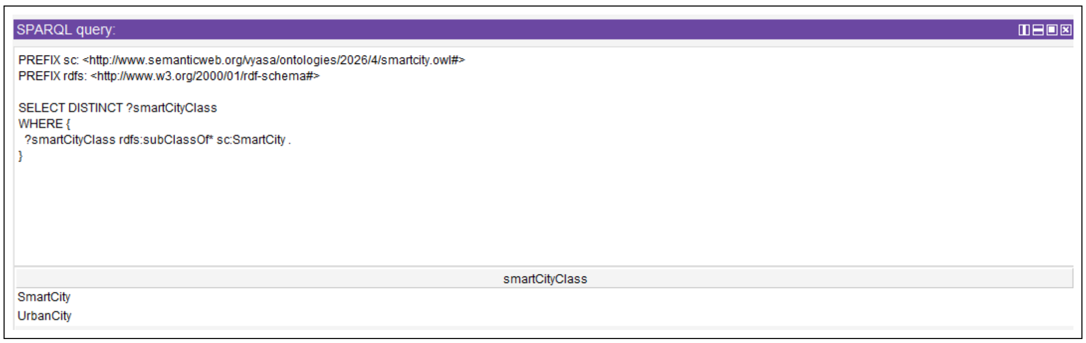


Figure 5: SPARQL query result for CQ1: returns the Smart City class and its subclasses.

CQ2: What are the primary domains of a Smart City?

```
SELECT DISTINCT ?domain
WHERE {
  ?domain rdfs:subClassOf ?restriction .
  ?restriction owl:onProperty sc:isADomainOf ;
               owl:someValuesFrom sc:SmartCity .
}
```

Listing 3: SPARQL query for CQ2 – primary domains

Result: The query returns `sc:SmartTransportation`, `sc:SmartEnergyGrid`, `sc:SmartWaterManagement`, and `sc:SmartHealthcare`, correctly identifying the four primary domains.

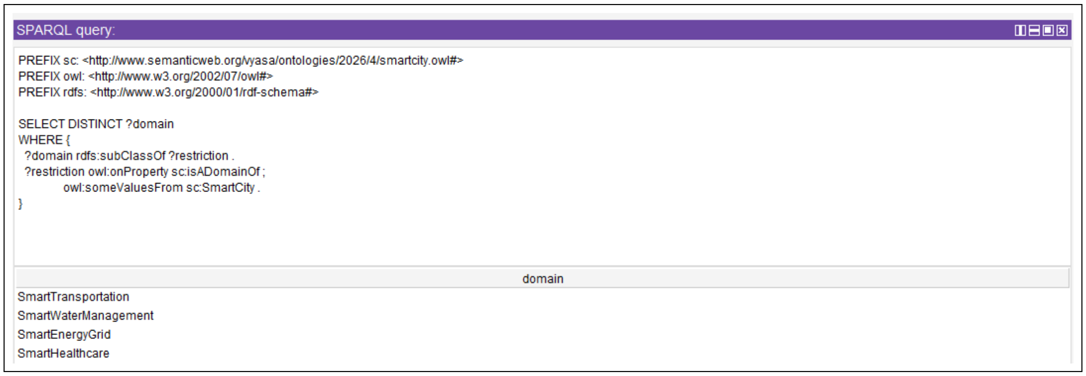


Figure 6: SPARQL query result for CQ2: returns the four primary Smart City domains.

CQ3: What technologies enable a Smart City?

```
SELECT DISTINCT ?technology
WHERE {
  VALUES ?technology {
    sc:IotPlatform sc:5gNetwork sc:EdgeComputingNode
    sc:CloudPlatform sc:DataAnalyticsEngine sc:DigitalTwin
    sc:CybersecuritySystem sc:UrbanSensorNetwork
  }
}
```

Listing 4: SPARQL query for CQ3 – enabling technologies

Result: The query returns the eight core enabling technologies defined in the ontology, including `sc:IotPlatform`, `sc:5gNetwork`, `sc:EdgeComputingNode`, `sc:CloudPlatform`, `sc:DataAnalyticsEngine`, `sc:DigitalTwin`, `sc:CybersecuritySystem`, and `sc:UrbanSensorNetwork`.

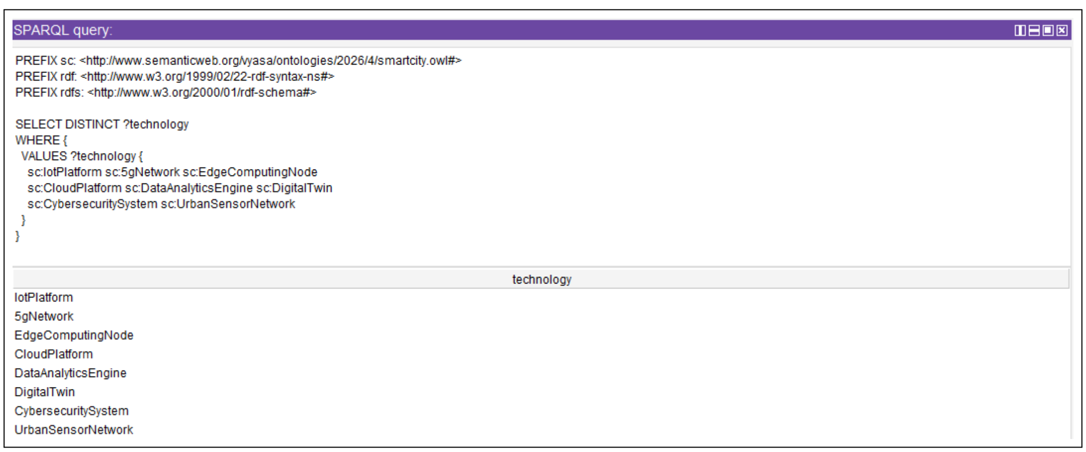


Figure 7: SPARQL query result for CQ3: returns the enabling technologies of a Smart City.

CQ4: What is Smart Transportation?

```
SELECT DISTINCT ?domain ?property ?value
WHERE {
  sc:SmartTransportation rdfs:subClassOf ?restriction .
  ?restriction owl:onProperty ?property ;
```

```
                    owl:someValuesFrom ?value .
}
```

Listing 5: SPARQL query for CQ4 – Smart Transportation definition

Result: The query returns the object properties and range values that define `sc:SmartTransportation` through its OWL existential restrictions, capturing the semantic definition of this domain.

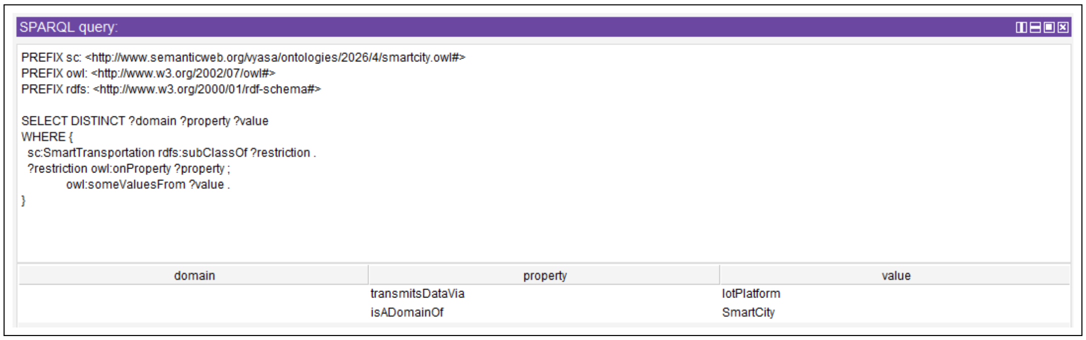


Figure 8: SPARQL query result for CQ4: returns the defining properties of Smart Transportation.

CQ5: What are the components of a Smart Transportation system?

```
SELECT DISTINCT ?component
WHERE {
  ?component rdfs:subClassOf ?restriction .
  ?restriction owl:onProperty sc:isAComponentOf ;
               owl:someValuesFrom sc:SmartTransportation .
}
```

Listing 6: SPARQL query for CQ5 – Smart Transportation components

Result: The query returns all classes that are components of `sc:SmartTransportation` via the `sc:isAComponentOf` property, such as traffic management systems, public transit systems etc.

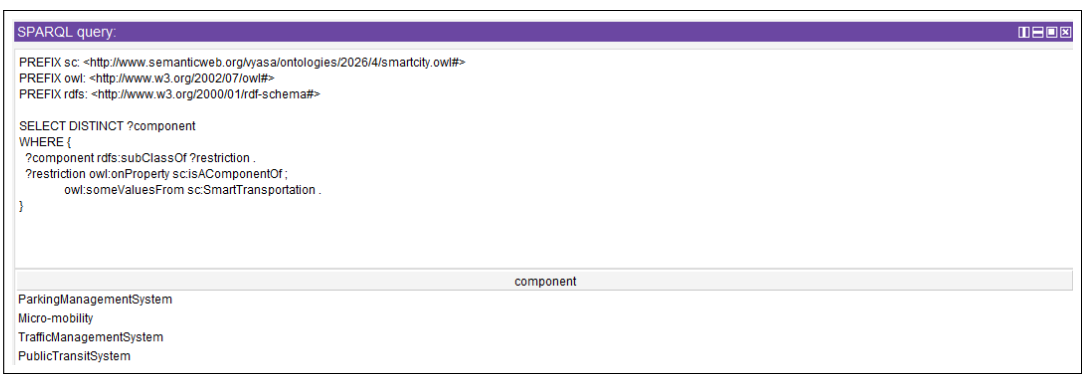


Figure 9: SPARQL query result for CQ5: returns the components of the Smart Transportation system.

CQ6: How does traffic management work in a Smart City?

```
SELECT DISTINCT ?component ?property ?value
WHERE {
  ?component rdfs:subClassOf ?r1 .
```

```
  ?r1 owl:onProperty sc:isAComponentOf ;
      owl:someValuesFrom sc:TrafficManagementSystem .
  ?component rdfs:subClassOf ?r2 .
  ?r2 owl:onProperty ?property ;
      owl:someValuesFrom ?value .
}
```

Listing 7: SPARQL query for CQ6 – traffic management

Result: The query returns the subcomponents of `sc:TrafficManagementSystem` together with their associated properties and values, illustrating how traffic management is modelled in the ontology.

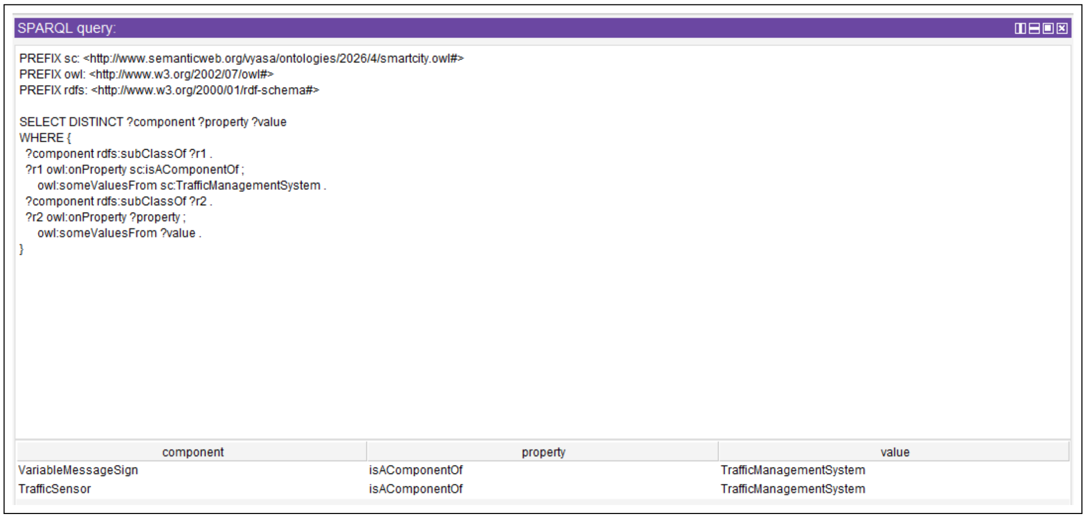


Figure 10: SPARQL query result for CQ6: returns traffic management components and their properties.

CQ7: What types of smart mobility solutions exist?

```
SELECT DISTINCT ?mobilityInstance
WHERE {
  { ?mobilityInstance rdf:type sc:Micro-mobility }
  UNION
  { ?mobilityInstance rdf:type sc:ConnectedVehicle }
}
```

Listing 8: SPARQL query for CQ7 – smart mobility solutions

Result: The query returns all named individuals that are instances of `sc:Micro-mobility` or `sc:ConnectedVehicle`, covering the range of smart mobility solutions represented in the ontology.

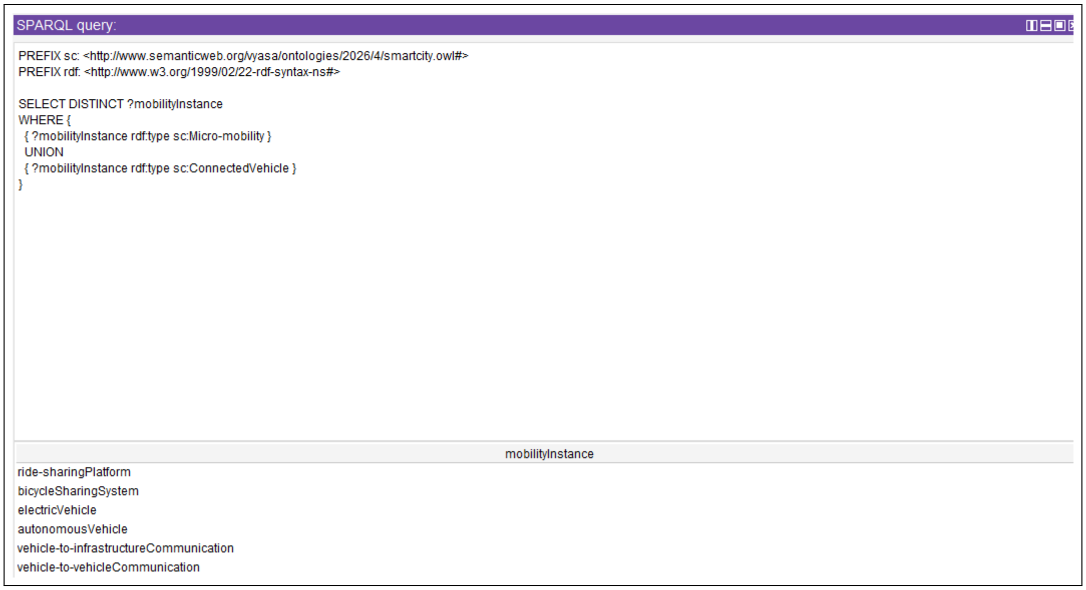


Figure 11: SPARQL query result for CQ7: returns instances of smart mobility solution types.

CQ8: How is public transit integrated in a Smart City?

```
SELECT DISTINCT ?transitInstance ?component
WHERE {
  ?transitInstance rdf:type sc:PublicTransitSystem .
  ?component rdfs:subClassOf ?r .
  ?r owl:onProperty sc:isAComponentOf ;
     owl:someValuesFrom sc:PublicTransitSystem .
}
```

Listing 9: SPARQL query for CQ8 – public transit integration

Result: The query returns instances of `sc:PublicTransitSystem` alongside the classes that serve as its components, demonstrating how public transit is structurally integrated within the Smart Transportation domain.

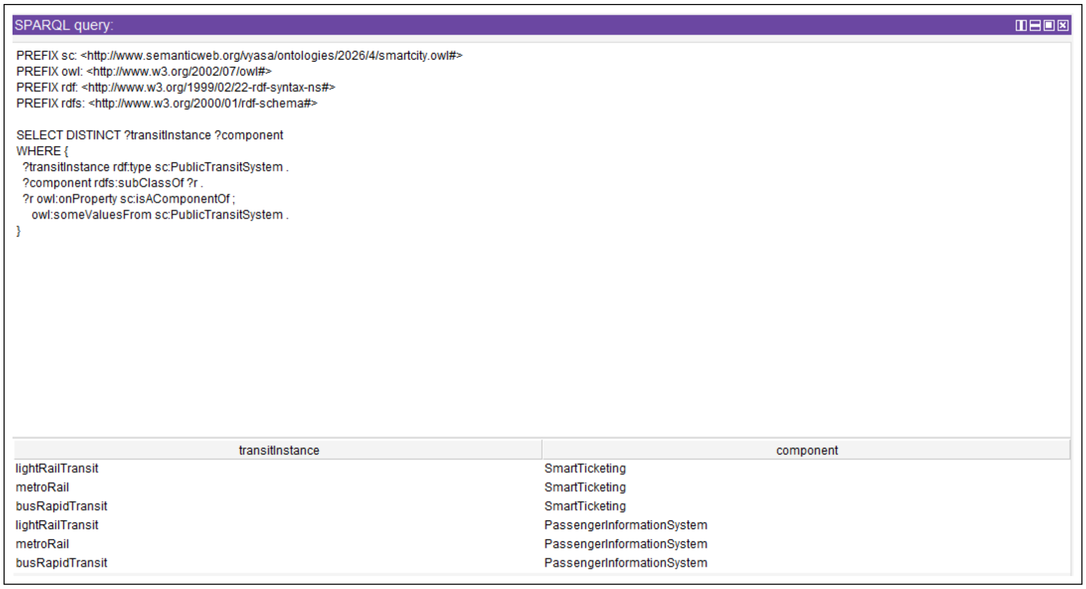


Figure 12: SPARQL query result for CQ8: returns public transit instances and their subcomponents.

CQ9: How are electric vehicles supported in a Smart City?

```
SELECT DISTINCT ?ev ?chargingStation
WHERE {
  ?ev rdf:type sc:ConnectedVehicle .
  ?ev sc:uses ?chargingStation .
}
```

Listing 10: SPARQL query for CQ9 – electric vehicle support

Result: The query returns pairs of connected vehicle instances and the charging stations they use via the `sc:uses` property, capturing how electric vehicle infrastructure is represented in the ontology.

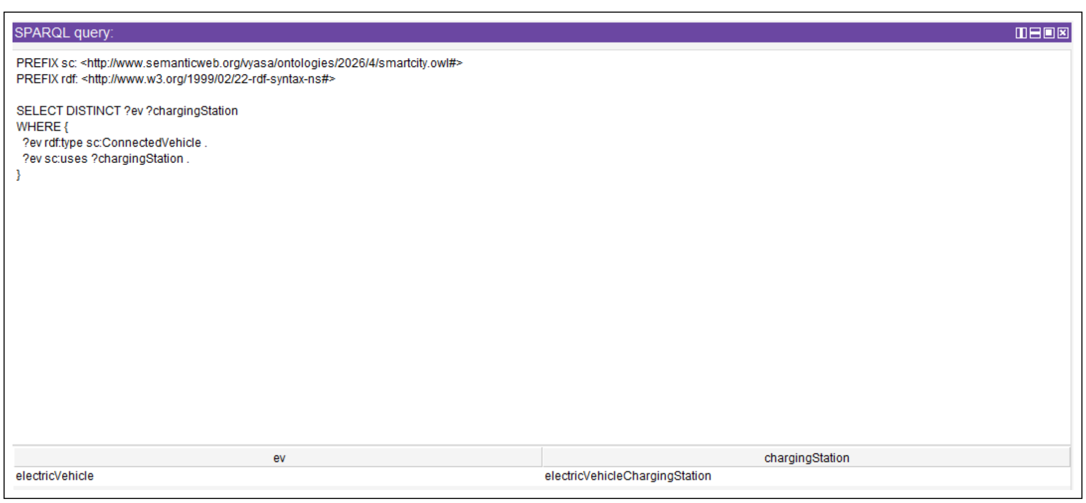


Figure 13: SPARQL query result for CQ9: returns electric vehicles and their associated charging stations.

CQ10: What is a Smart Energy Grid?

```
SELECT DISTINCT ?property ?value
WHERE {
  sc:SmartEnergyGrid rdfs:subClassOf ?restriction .
  ?restriction owl:onProperty ?property ;
               owl:someValuesFrom ?value .
}
```

Listing 11: SPARQL query for CQ10 – Smart Energy Grid definition

Result: The query returns the object properties and their range values associated with `sc:SmartEnergyGrid`, capturing the ontological definition of the Smart Energy Grid domain.

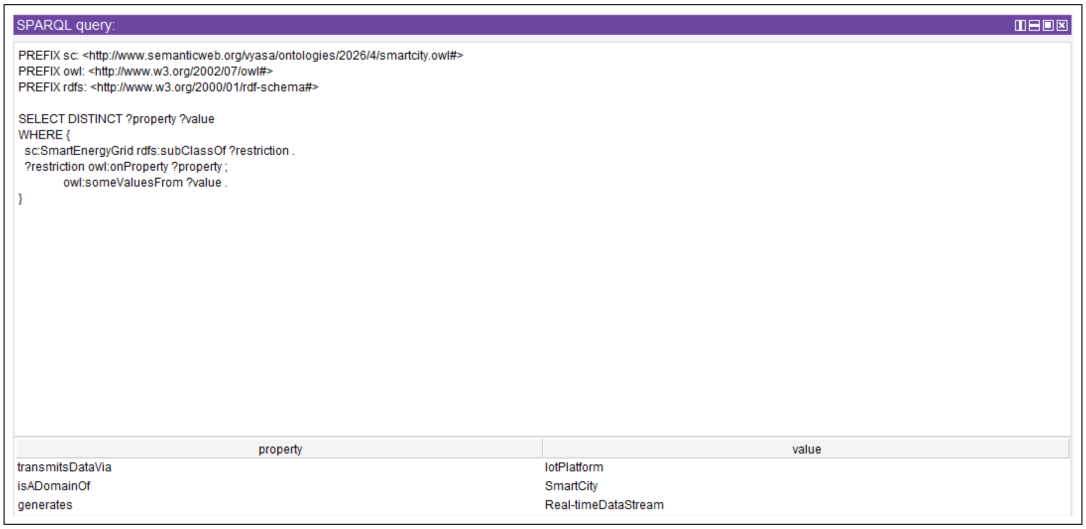


Figure 14: SPARQL query result for CQ10: returns the defining properties of the Smart Energy Grid.

CQ11: What are the components of a Smart Grid?

```
SELECT DISTINCT ?component
WHERE {
  ?component rdfs:subClassOf ?restriction .
  ?restriction owl:onProperty sc:isAComponentOf ;
               owl:someValuesFrom sc:SmartEnergyGrid .
}
```

Listing 12: SPARQL query for CQ11 – Smart Grid components

Result: The query returns all classes that are declared as components of `sc:SmartEnergyGrid` via `sc:isAComponentOf`, such as sub stations, distributed energy resources, and load balancing system and advanced metering infrastructure.

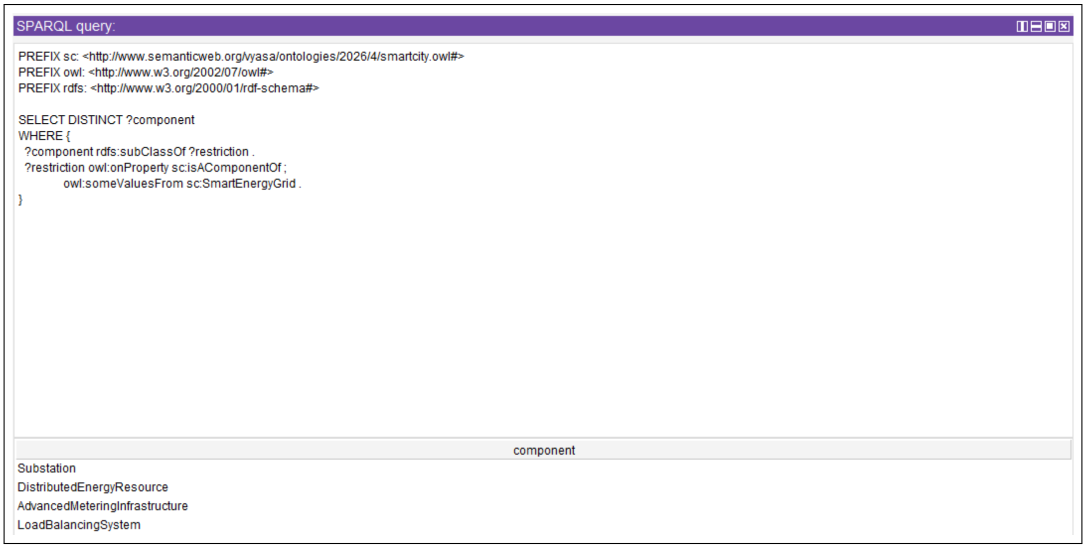


Figure 15: SPARQL query result for CQ11: returns the components of the Smart Energy Grid.

CQ12: How is energy consumption monitored?

```
SELECT DISTINCT ?monitor ?target
WHERE {
  ?monitor rdfs:subClassOf ?restriction .
  ?restriction owl:onProperty sc:monitors ;
               owl:someValuesFrom ?target .
  FILTER(?target = sc:EnergyConsumption)
}
```

Listing 13: SPARQL query for CQ12 – energy consumption monitoring

Result: The query returns the classes that monitor `sc:EnergyConsumption` via the `sc:monitors` property, identifying the systems responsible for energy consumption tracking in the ontology.

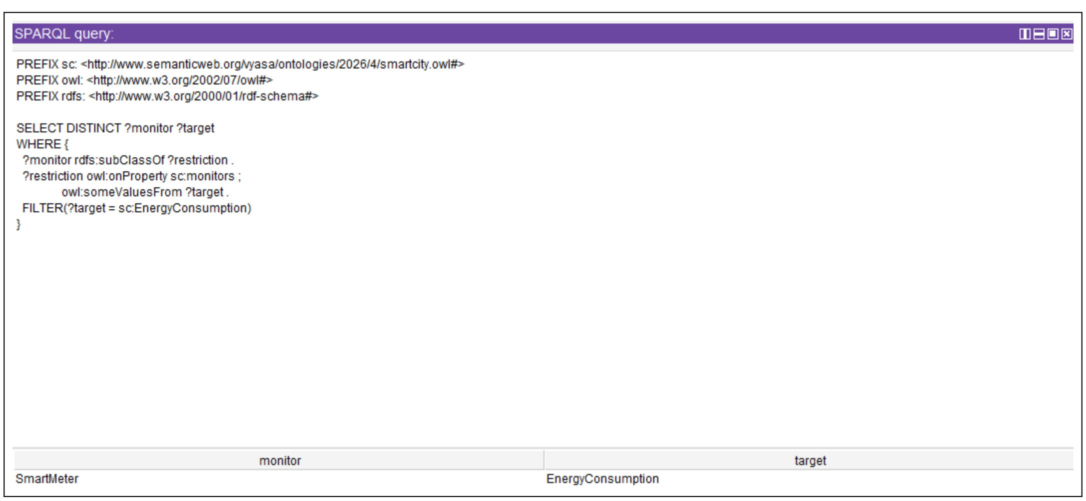


Figure 16: SPARQL query result for CQ12: returns the systems that monitor energy consumption.

CQ13: How is renewable energy integrated?

```
SELECT DISTINCT ?resource ?integratedBy
WHERE {
  ?resource rdf:type sc:DistributedEnergyResource .
  ?integratedBy rdfs:subClassOf ?r .
  ?r owl:onProperty sc:integrates ;
     owl:someValuesFrom sc:DistributedEnergyResource .
}
```

Listing 14: SPARQL query for CQ13 – renewable energy integration

Result: The query returns instances of `sc:DistributedEnergyResource` alongside the classes that integrate them, demonstrating how renewable energy sources are incorporated into the Smart Energy Grid.

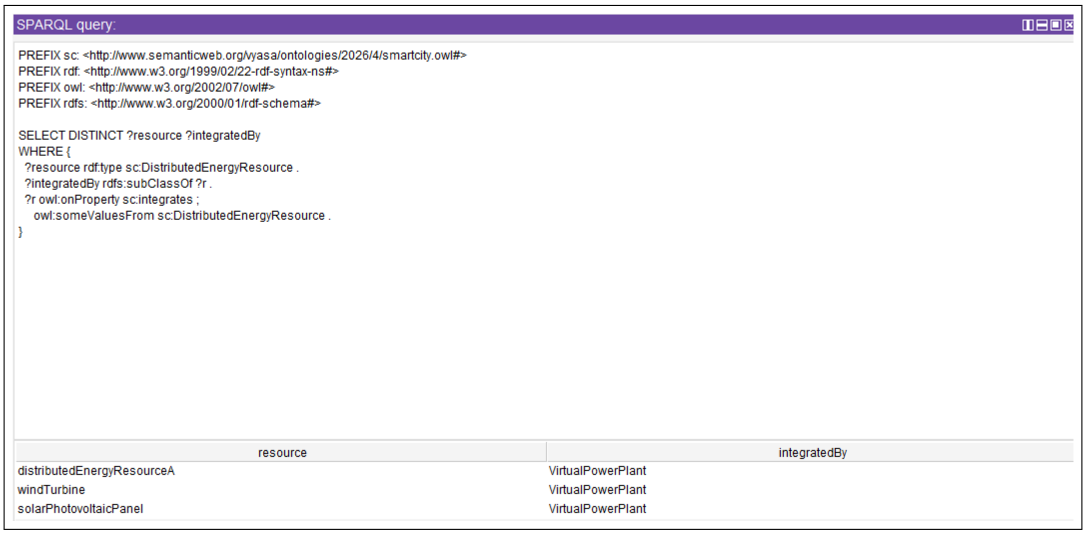


Figure 17: SPARQL query result for CQ13: returns renewable energy resources and the systems that integrate them.

CQ14: How does demand response work?

```
SELECT DISTINCT ?system ?property ?value
WHERE {
  sc:DemandResponseSystem rdfs:subClassOf ?restriction .
  ?restriction owl:onProperty ?property ;
               owl:someValuesFrom ?value .
  ?system rdfs:subClassOf ?r2 .
  ?r2 owl:onProperty sc:controls ;
      owl:someValuesFrom sc:DemandResponseSystem .
}
```

Listing 15: SPARQL query for CQ14 – demand response

Result: The query returns the properties defining `sc:DemandResponseSystem` as well as the classes that control it, capturing the demand response mechanism as modelled in the ontology.

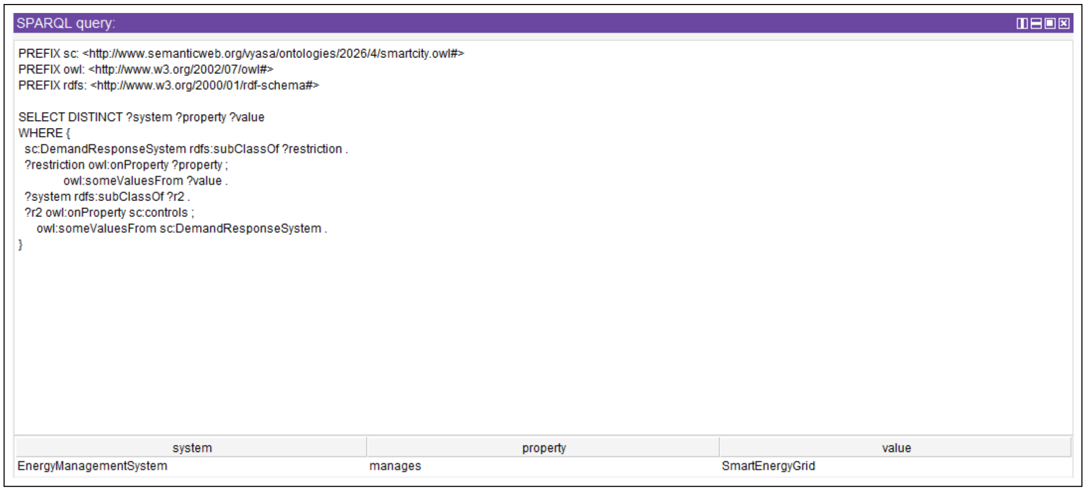


Figure 18: SPARQL query result for CQ14: returns the demand response system properties and controlling classes.

CQ15: What is Smart Water Management?

```
SELECT DISTINCT ?property ?value
WHERE {
  sc:SmartWaterManagement rdfs:subClassOf ?restriction .
  ?restriction owl:onProperty ?property ;
               owl:someValuesFrom ?value .
}
```

Listing 16: SPARQL query for CQ15 – Smart Water Management definition

Result: The query returns the object properties and their range values associated with `sc:SmartWaterManagement`, capturing the ontological definition of this domain.

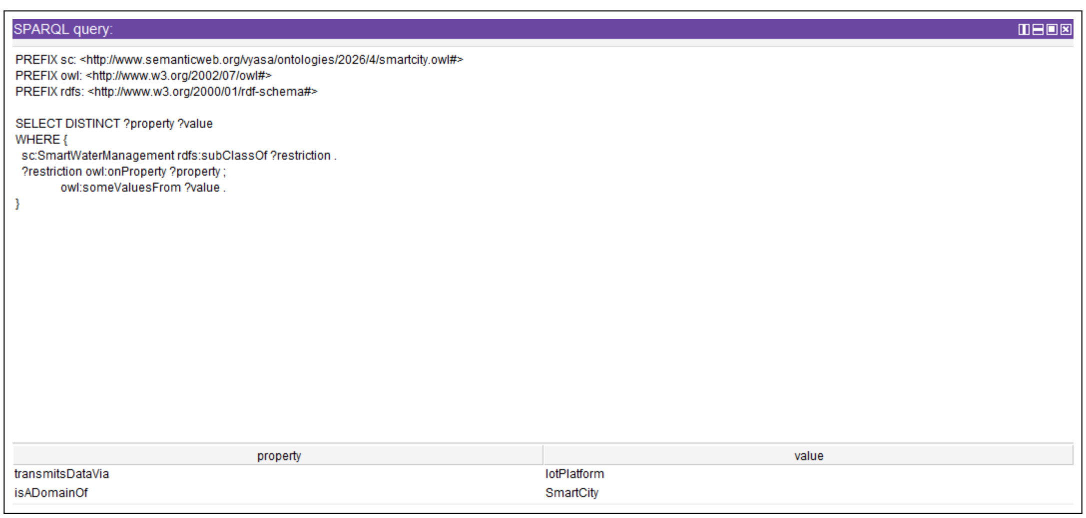


Figure 19: SPARQL query result for CQ15: returns the defining properties of Smart Water Management.

CQ16: What are the components of a Smart Water system?

```
SELECT DISTINCT ?component
WHERE {
  ?component rdfs:subClassOf ?restriction .
  ?restriction owl:onProperty sc:isAComponentOf ;
               owl:someValuesFrom sc:SmartWaterManagement .
}
```

Listing 17: SPARQL query for CQ16 – Smart Water system components

Result: The query returns all classes that are declared components of `sc:SmartWaterManagement`, including Water Treatment Plants, reservoir, Stormwater Management System , Water Distribution Network.

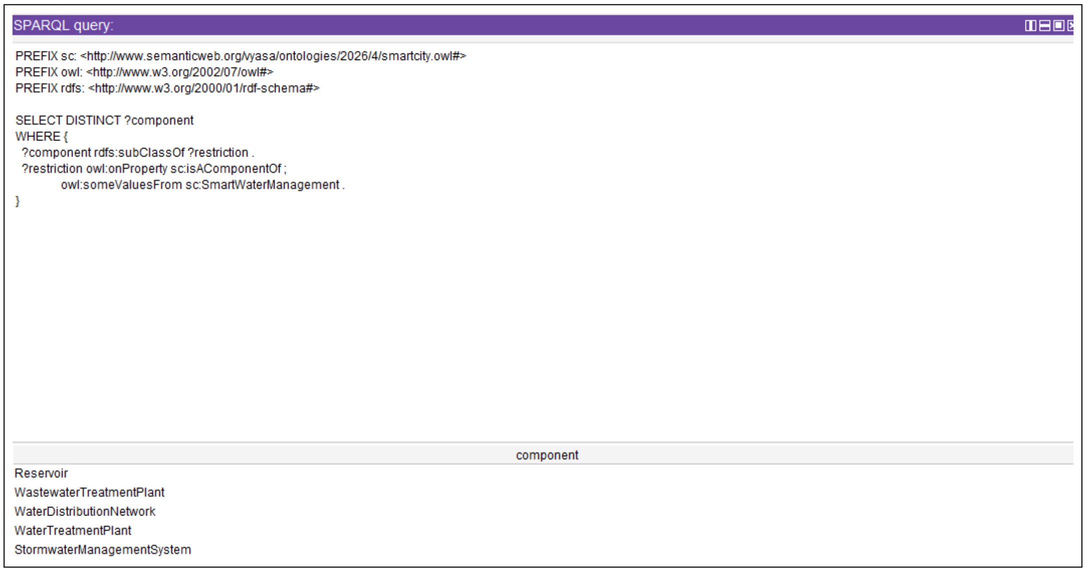


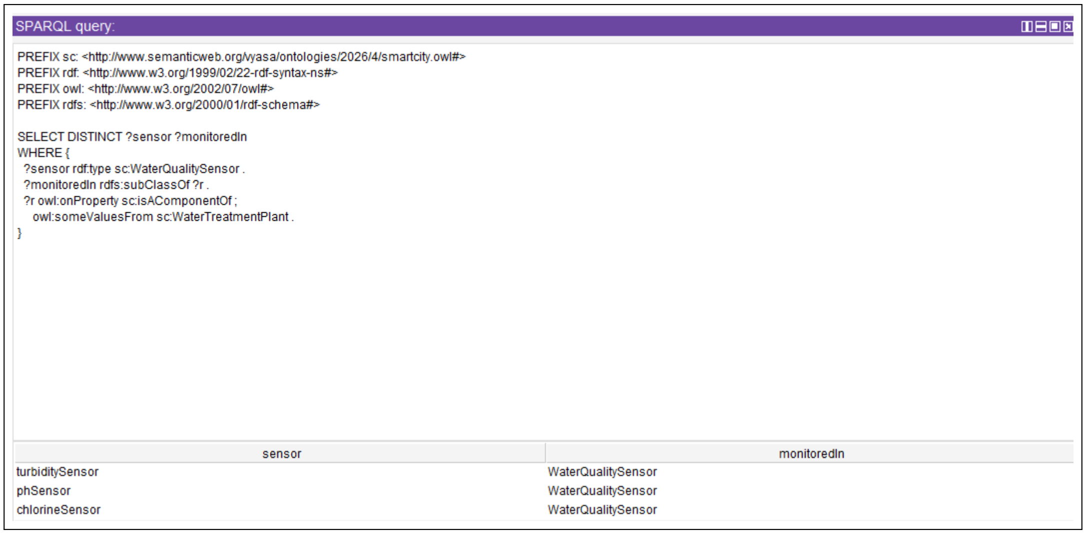
Figure 20: SPARQL query result for CQ16: returns the components of the Smart Water Management system.

CQ17: How is water quality monitored?

```
SELECT DISTINCT ?sensor ?monitoredIn
WHERE {
  ?sensor rdf:type sc:WaterQualitySensor .
  ?monitoredIn rdfs:subClassOf ?r .
  ?r owl:onProperty sc:isAComponentOf ;
     owl:someValuesFrom sc:WaterTreatmentPlant .
}
```

Listing 18: SPARQL query for CQ17 – water quality monitoring

Result: The query returns instances of `sc:WaterQualitySensor` and the classes that are components of `sc:WaterTreatmentPlant`, showing how water quality monitoring is represented in the ontology.



SPARQL query:

```
PREFIX sc: <http://www.semanticweb.org/vyasa/ontologies/2026/4/smartcity.owl#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?sensor ?monitoredIn
WHERE {
 ?sensor rdf:type sc:WaterQualitySensor .
 ?monitoredIn rdfs:subClassOf ?r .
 ?r owl:onProperty sc:isAComponentOf ;
   owl:someValuesFrom sc:WaterTreatmentPlant .
}
```

sensor	monitoredIn
turbiditySensor	WaterQualitySensor
phSensor	WaterQualitySensor
chlorineSensor	WaterQualitySensor

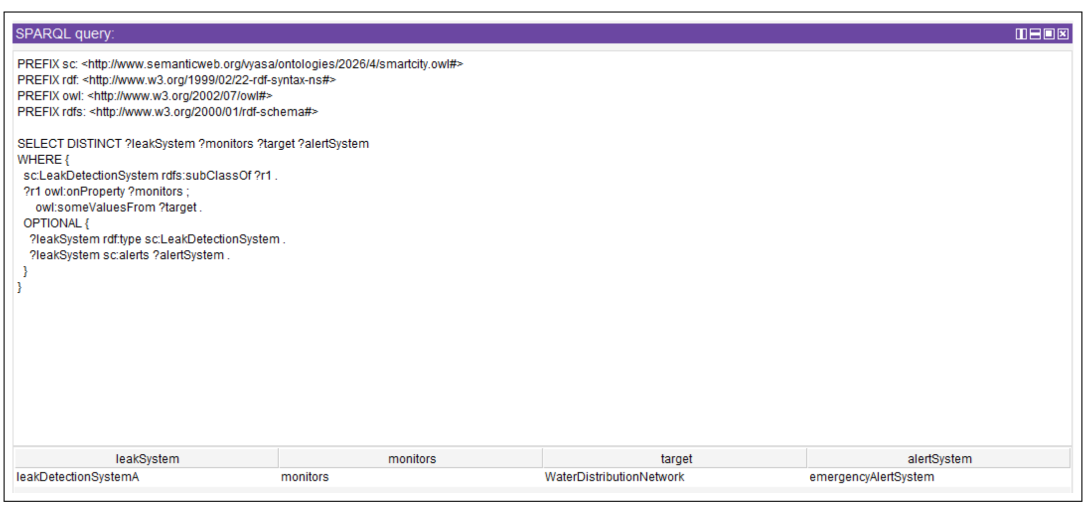
Figure 21: SPARQL query result for CQ17: returns water quality sensors and their monitoring context.

CQ18: How does leak detection work?

```
SELECT DISTINCT ?leakSystem ?monitors ?target ?alertSystem
WHERE {
  sc:LeakDetectionSystem rdfs:subClassOf ?r1 .
  ?r1 owl:onProperty ?monitors ;
      owl:someValuesFrom ?target .
  OPTIONAL {
    ?leakSystem rdf:type sc:LeakDetectionSystem .
    ?leakSystem sc:alerts ?alertSystem .
  }
}
```

Listing 19: SPARQL query for CQ18 – leak detection

Result: The query returns the properties and monitored targets of `sc:LeakDetectionSystem`, as well as any alert relationships to other systems, capturing the complete leak detection workflow.



SPARQL query:

```
PREFIX sc: <http://www.semanticweb.org/vyasa/ontologies/2026/4/smartcity.owl#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?leakSystem ?monitors ?target ?alertSystem
WHERE {
 sc:LeakDetectionSystem rdfs:subClassOf ?r1 .
 ?r1 owl:onProperty ?monitors ;
   owl:someValuesFrom ?target .
 OPTIONAL {
  ?leakSystem rdf:type sc:LeakDetectionSystem .
  ?leakSystem sc:alerts ?alertSystem .
 }
}
```

leakSystem	monitors	target	alertSystem
leakDetectionSystemA	monitors	WaterDistributionNetwork	emergencyAlertSystem

Figure 22: SPARQL query result for CQ18: returns leak detection system properties and alert targets.

CQ19: How is wastewater treated and recycled?

```
SELECT DISTINCT ?plant ?output
WHERE {
  ?plant rdf:type sc:WastewaterTreatmentPlant .
  ?plant sc:produces ?output .
}
```

Listing 20: SPARQL query for CQ19 – wastewater treatment and recycling

Result: The query returns instances of `sc:WastewaterTreatmentPlant` and the outputs they produce via the `sc:produces` property, representing how treated water and recycled resources are modelled.

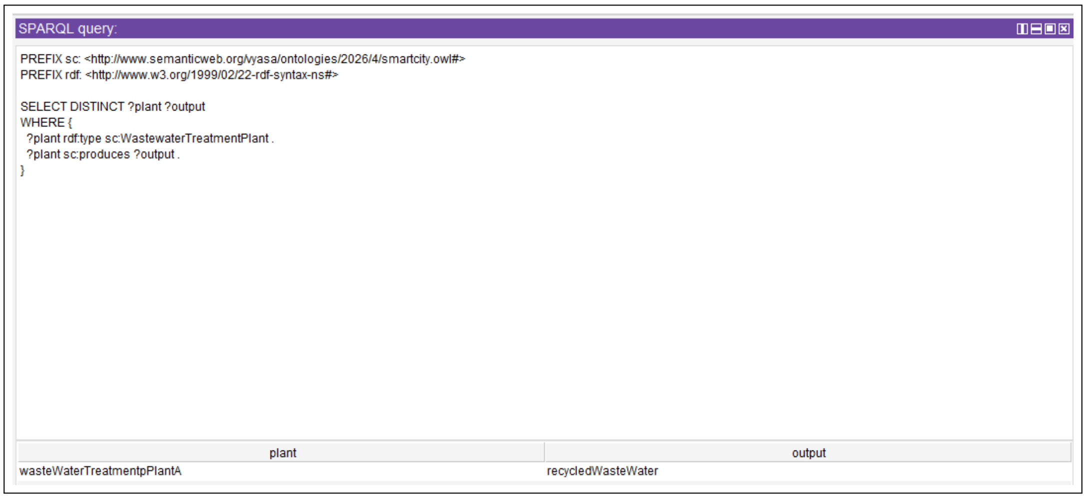


Figure 23: SPARQL query result for CQ19: returns wastewater treatment plants and their produced outputs.

CQ20: What is Smart Healthcare?

```
SELECT DISTINCT ?property ?value
WHERE {
  sc:SmartHealthcare rdfs:subClassOf ?restriction .
  ?restriction owl:onProperty ?property ;
               owl:someValuesFrom ?value .
}
```

Listing 21: SPARQL query for CQ20 – Smart Healthcare definition

Result: The query returns the object properties and range values that define `sc:SmartHealthcare` through its OWL existential restrictions.

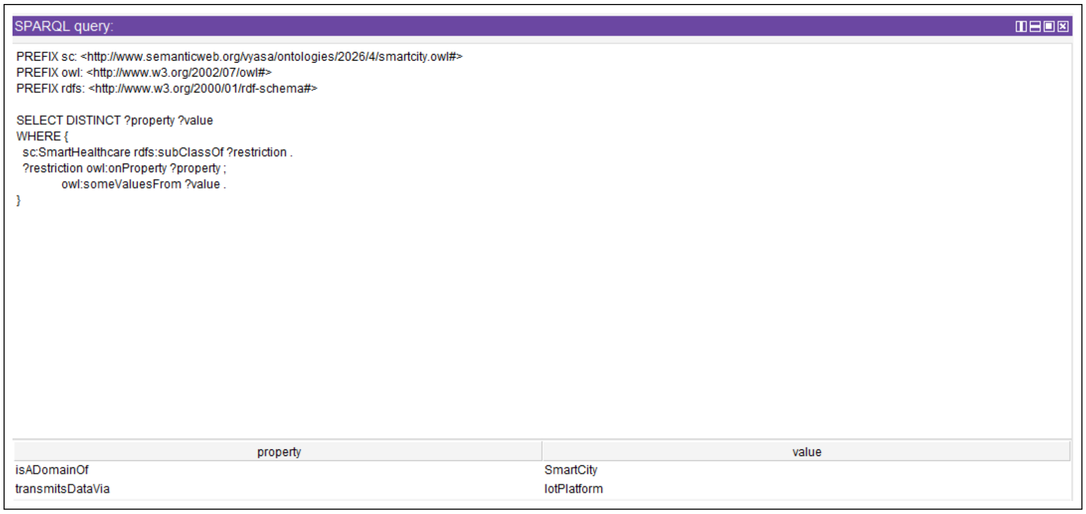


Figure 24: SPARQL query result for CQ20: returns the defining properties of Smart Healthcare.

CQ21: What are the components of a Smart Healthcare system?

```
SELECT DISTINCT ?component
WHERE {
  ?component rdfs:subClassOf ?restriction .
  ?restriction owl:onProperty sc:isAComponentOf ;
               owl:someValuesFrom sc:SmartHealthcare .
}
```

Listing 22: SPARQL query for CQ21 – Smart Healthcare components

Result: The query returns all classes declared as components of `sc:SmartHealthcare`, such as Electronic Health Record,Smart Pharmacy and Ambulance Dispatch System.

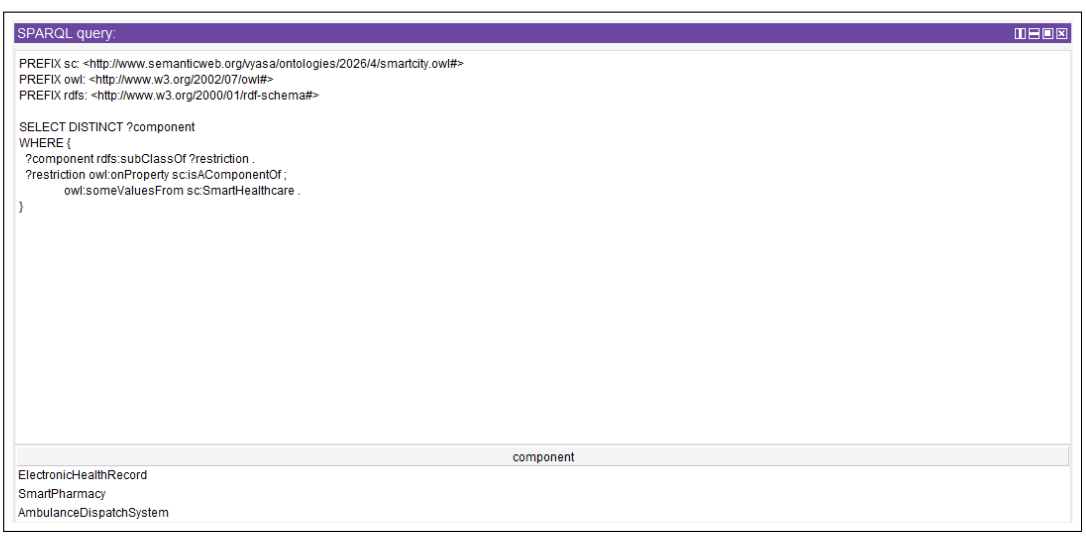


Figure 25: SPARQL query result for CQ21: returns the components of the Smart Healthcare system.

CQ22: How are health records managed?

```
SELECT DISTINCT ?system ?property ?target
WHERE {
  ?system rdfs:subClassOf ?r .
  ?r owl:onProperty ?property ;
     owl:someValuesFrom sc:ElectronicHealthRecord .
}
```

Listing 23: SPARQL query for CQ22 – health record management

Result: The query returns the classes that interact with `sc:ElectronicHealthRecord` via any object property, identifying all systems involved in managing electronic health records.

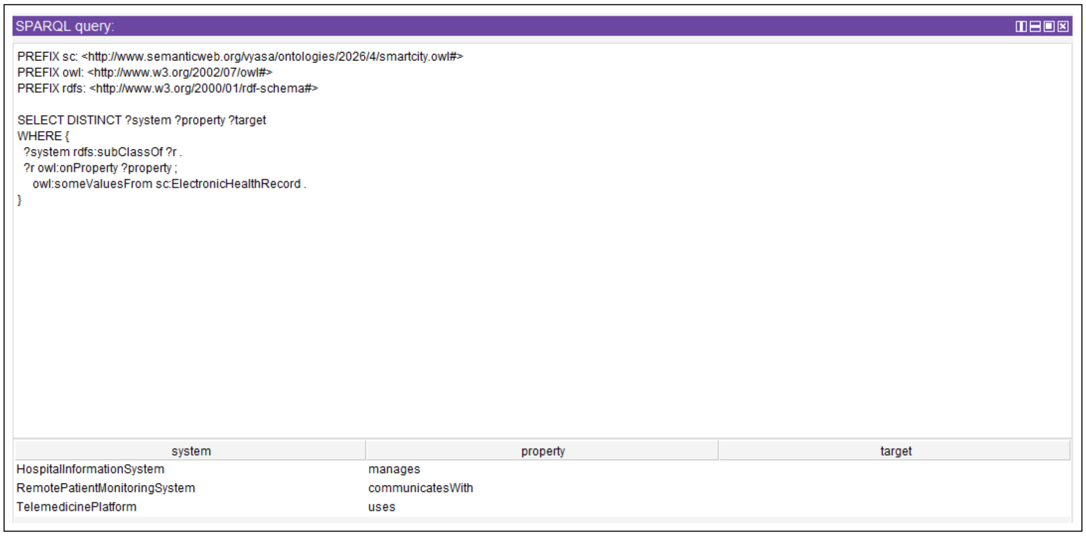


Figure 26: SPARQL query result for CQ22: returns systems that manage electronic health records.

CQ23: What wearable devices are used?

```
SELECT DISTINCT ?wearable
WHERE {
  ?wearable rdf:type sc:WearableHealthSensor .
}
```

Listing 24: SPARQL query for CQ23 – wearable health devices

Result: The query returns all named individuals that are instances of `sc:WearableHealthSensor`, enumerating the wearable devices represented in the ontology.

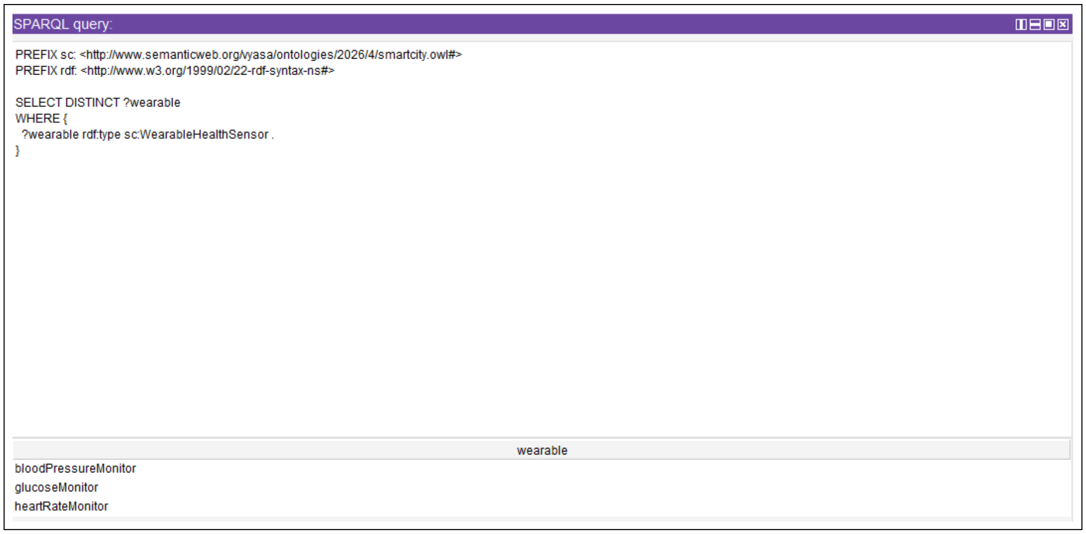


Figure 27: SPARQL query result for CQ23: returns wearable health sensor instances.

CQ24: How does telemedicine function?

```
SELECT DISTINCT ?property ?value
WHERE {
  sc:TelemedicinePlatform rdfs:subClassOf ?restriction .
  ?restriction owl:onProperty ?property ;
               owl:someValuesFrom ?value .
}
```

Listing 25: SPARQL query for CQ24 – telemedicine functionality

Result: The query returns the object properties and range values associated with `sc:TelemedicinePlatform`, capturing the functional relationships that define telemedicine in the ontology.

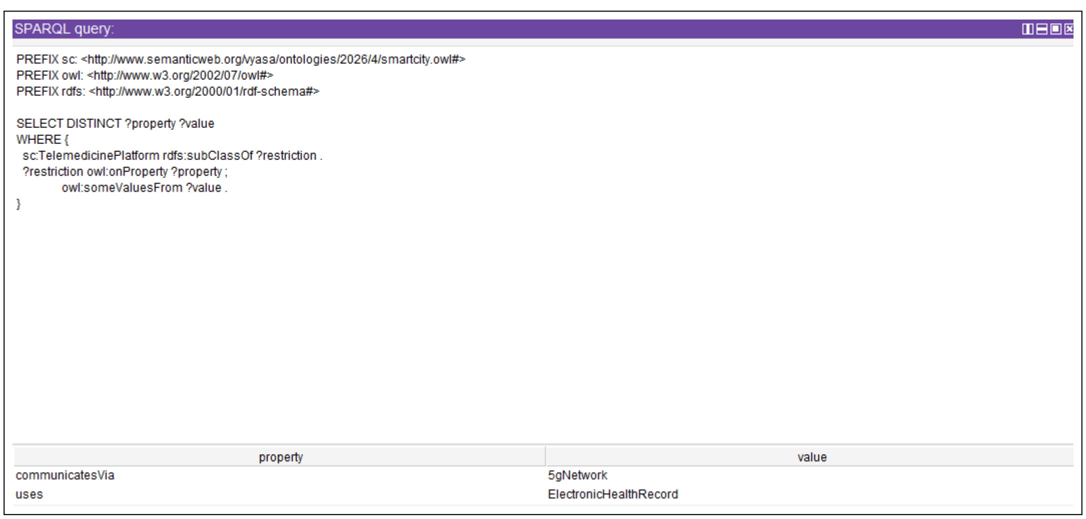


Figure 28: SPARQL query result for CQ24: returns the functional properties of the Telemedicine Platform.

CQ25: How are ambulances managed?

```
SELECT DISTINCT ?ambulance ?communicatesWith ?system
WHERE {
  ?ambulance rdf:type sc:AmbulanceDispatchSystem .
  OPTIONAL { ?ambulance sc:communicatesWith ?system }
  ?component rdfs:subClassOf ?r .
  ?r owl:onProperty sc:isAComponentOf ;
     owl:someValuesFrom sc:SmartHealthcare .
  BIND(sc:communicatesWith AS ?communicatesWith)
}
```

Listing 26: SPARQL query for CQ25 – ambulance management

Result: The query returns instances of `sc:AmbulanceDispatchSystem`, their communication targets via `sc:communicatesWith`, and the broader healthcare component context, representing ambulance management in the ontology.

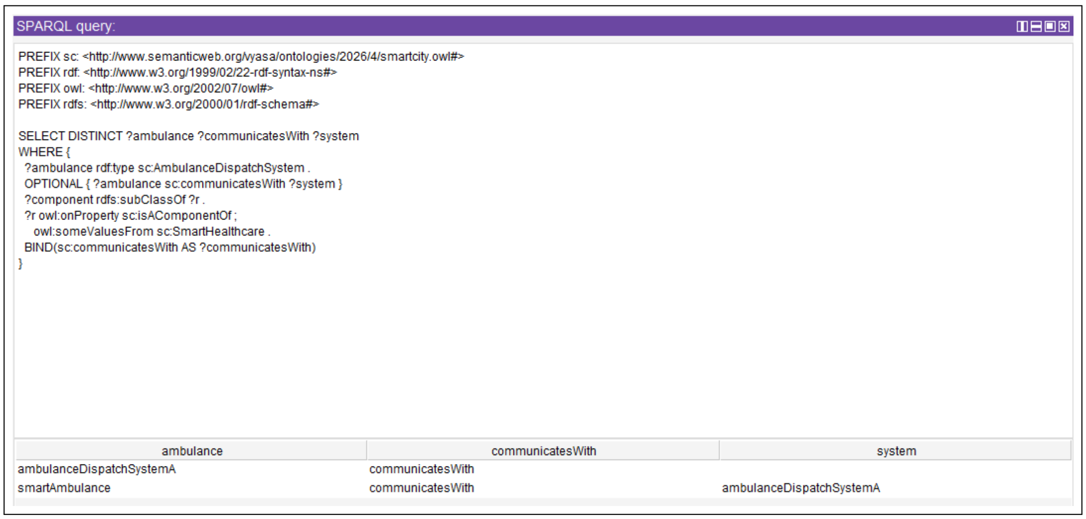


Figure 29: SPARQL query result for CQ25: returns ambulance dispatch systems and their communication relationships.

CQ26: What is the IoT Platform's role?

```
SELECT DISTINCT ?domain
WHERE {
  ?domain rdfs:subClassOf ?restriction .
  ?restriction owl:onProperty sc:transmitsDataVia ;
               owl:someValuesFrom sc:IotPlatform .
}
```

Listing 27: SPARQL query for CQ26 – IoT Platform role

Result: The query returns all domains that transmit data via `sc:IotPlatform`, illustrating the central role of the IoT Platform as the data transmission backbone across Smart City domains.

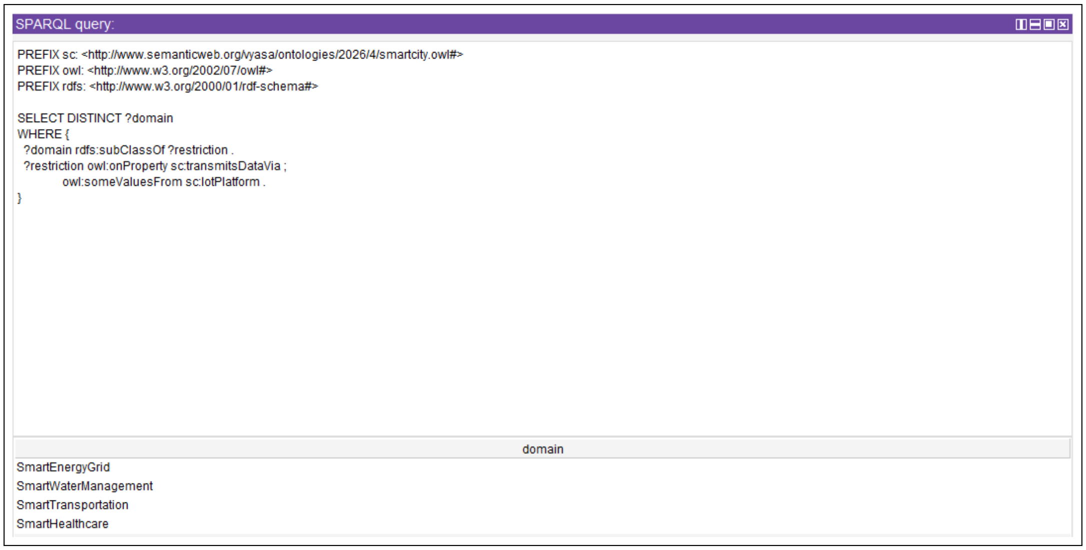


Figure 30: SPARQL query result for CQ26: returns domains that use the IoT Platform for data transmission.

CQ27: What types of sensors are used?

```
SELECT DISTINCT ?sensor ?type
WHERE {
  { ?sensor rdf:type sc:TrafficSensor .
    BIND("TrafficSensor" AS ?type) }
  UNION
  { ?sensor rdf:type sc:WaterQualitySensor .
    BIND("WaterQualitySensor" AS ?type) }
  UNION
  { ?sensor rdf:type sc:WearableHealthSensor .
    BIND("WearableHealthSensor" AS ?type) }
}
```

Listing 28: SPARQL query for CQ27 – sensor types

Result: The query returns all sensor instances across three sensor classes (`sc:TrafficSensor`, `sc:WaterQualitySensor`, `sc:WearableHealthSensor`), together with their type label.

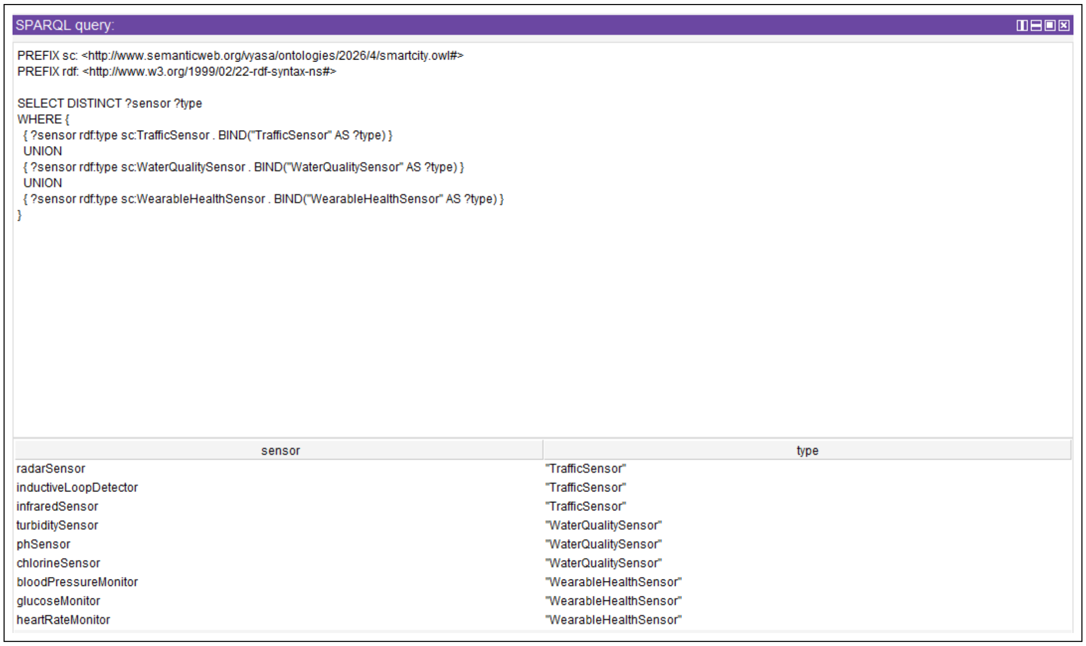


Figure 31: SPARQL query result for CQ27: returns all sensor instances categorised by type.

CQ28: What is a Digital Twin?

```
SELECT DISTINCT ?property ?value
WHERE {
  sc:DigitalTwin rdfs:subClassOf ?restriction .
  ?restriction owl:onProperty ?property ;
               owl:someValuesFrom ?value .
}
```

Listing 29: SPARQL query for CQ28 – Digital Twin definition

Result: The query returns the object properties and range values that define `sc:DigitalTwin` through its OWL existential restrictions, capturing its role and relationships within the Smart City ontology.

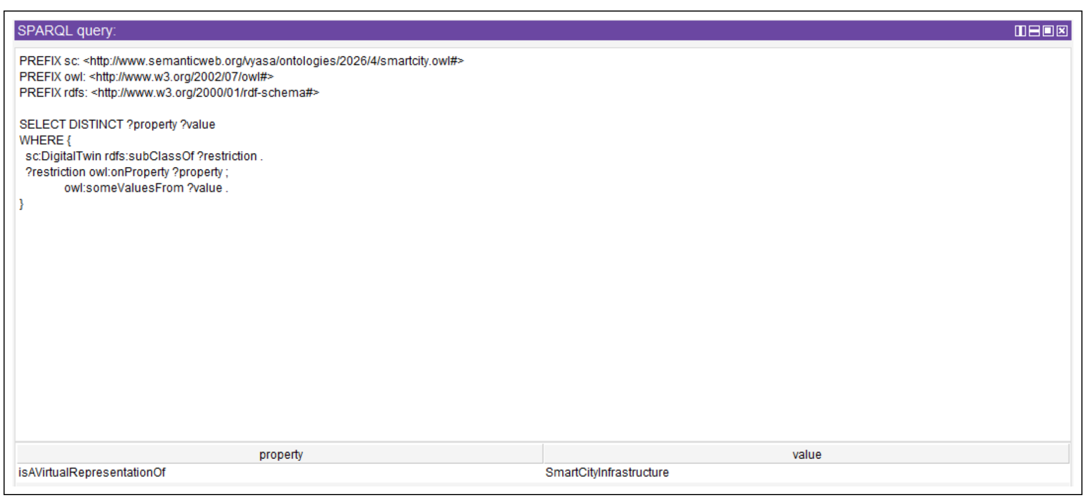


Figure 32: SPARQL query result for CQ28: returns the defining properties of the Digital Twin concept.

CQ29: How is data collected and processed?

```
SELECT ?source ?step1 ?step2 ?step3
WHERE {
  BIND(sc:UrbanSensorNetwork AS ?source)
  BIND(sc:IotPlatform AS ?step1)
  BIND(sc:EdgeComputingNode AS ?step2)
  BIND(sc:CloudPlatform AS ?step3)
  sc:IotPlatform rdfs:subClassOf ?r1 .
  ?r1 owl:onProperty sc:integrates .
  sc:EdgeComputingNode rdfs:subClassOf ?r2 .
  ?r2 owl:onProperty sc:processes .
  sc:CloudPlatform rdfs:subClassOf ?r3 .
  ?r3 owl:onProperty sc:stores .
}
```

Listing 30: SPARQL query for CQ29 – data collection and processing pipeline

Result: The query confirms the data pipeline from `sc:UrbanSensorNetwork` through `sc:IotPlatform` (integrates), `sc:EdgeComputingNode` (processes), and `sc:CloudPlatform` (stores), validating the four-stage data flow in the ontology.

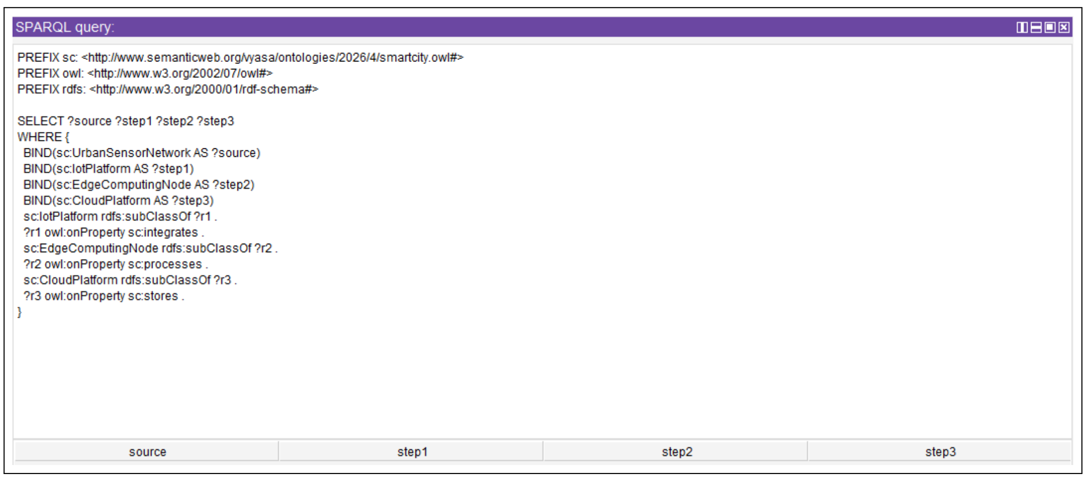


Figure 33: SPARQL query result for CQ29: returns the data collection and processing pipeline stages.

CQ30: What are the privacy and security concerns?

```
SELECT DISTINCT ?system ?property ?value
WHERE {
  { sc:CybersecuritySystem rdfs:subClassOf ?r .
    ?r owl:onProperty ?property ;
       owl:someValuesFrom ?value .
    BIND(sc:CybersecuritySystem AS ?system) }
  UNION
  { sc:PrivacyFramework rdfs:subClassOf ?r .
    ?r owl:onProperty ?property ;
       owl:someValuesFrom ?value .
    BIND(sc:PrivacyFramework AS ?system) }
}
```

Listing 31: SPARQL query for CQ30 – privacy and security

Result: The query returns the properties and associated values for both `sc:CybersecuritySystem` and `sc:PrivacyFramework`, exposing the security and privacy mechanisms modelled in the ontology.

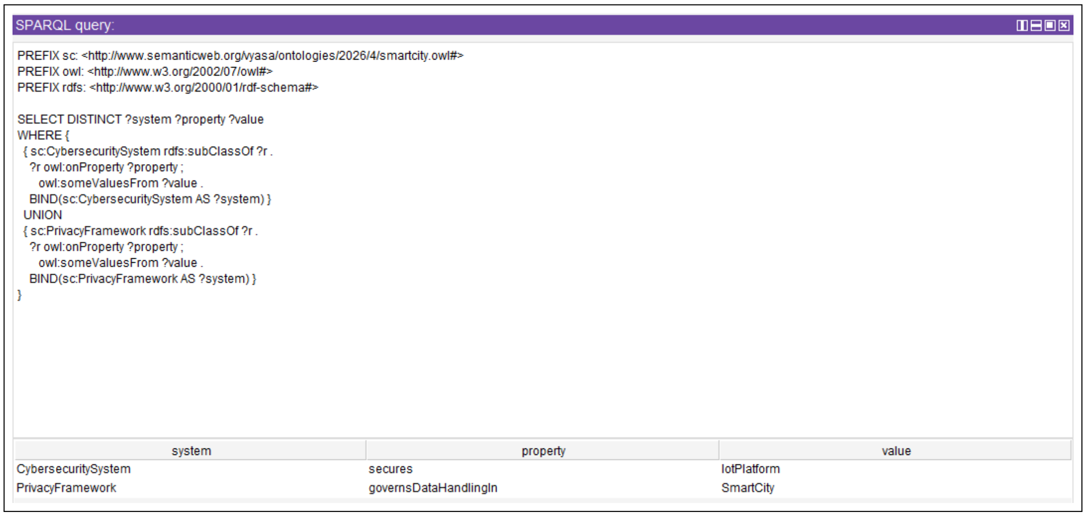


Figure 34: SPARQL query result for CQ30: returns the security and privacy system properties.

CQ31: What are the benefits of a Smart City?

```
SELECT DISTINCT ?system ?optimisesTarget
WHERE {
  ?system sc:optimises ?optimisesTarget .
}
```

Listing 32: SPARQL query for CQ31 – Smart City benefits

Result: The query returns all systems that optimise a target resource or process via the `sc:optimises` property, enumerating the concrete benefits represented in the ontology.

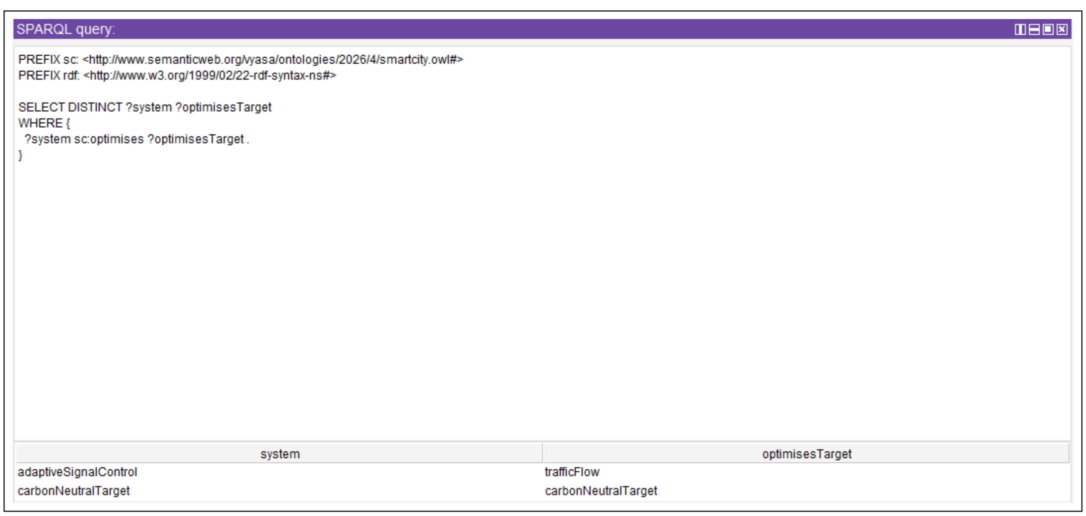


Figure 35: SPARQL query result for CQ31: returns systems and the targets they optimise.

CQ32: What are the challenges of implementing a Smart City?

```
SELECT DISTINCT ?challenge ?property ?value
WHERE {
  { sc:CybersecuritySystem rdfs:subClassOf ?r .
    ?r owl:onProperty ?property ;
       owl:someValuesFrom ?value .
    BIND(sc:CybersecuritySystem AS ?challenge) }
  UNION
  { sc:PrivacyFramework rdfs:subClassOf ?r .
    ?r owl:onProperty ?property ;
       owl:someValuesFrom ?value .
    BIND(sc:PrivacyFramework AS ?challenge) }
  UNION
  { ?challenge rdfs:subClassOf ?r2 .
    ?r2 owl:onProperty sc:isSecuredBy ;
        owl:someValuesFrom sc:CybersecuritySystem .
  }
}
```

Listing 33: SPARQL query for CQ32 – implementation challenges

Result: The query returns cybersecurity and privacy-related challenge classes alongside the domains that require securing, highlighting the implementation challenges modelled in the ontology.

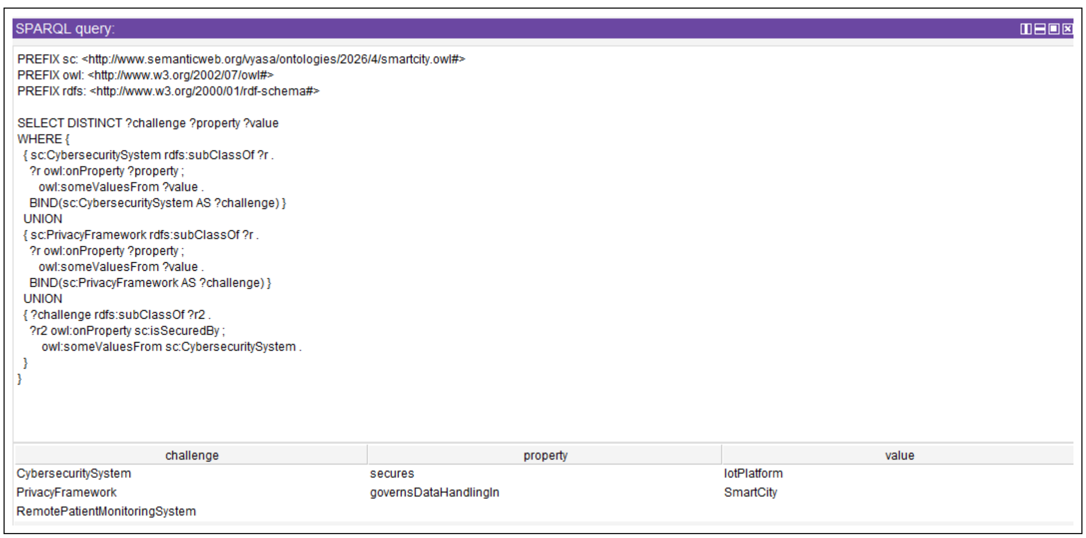


Figure 36: SPARQL query result for CQ32: returns implementation challenges related to security and privacy.

5.3.3 OWL Viz Class Diagram

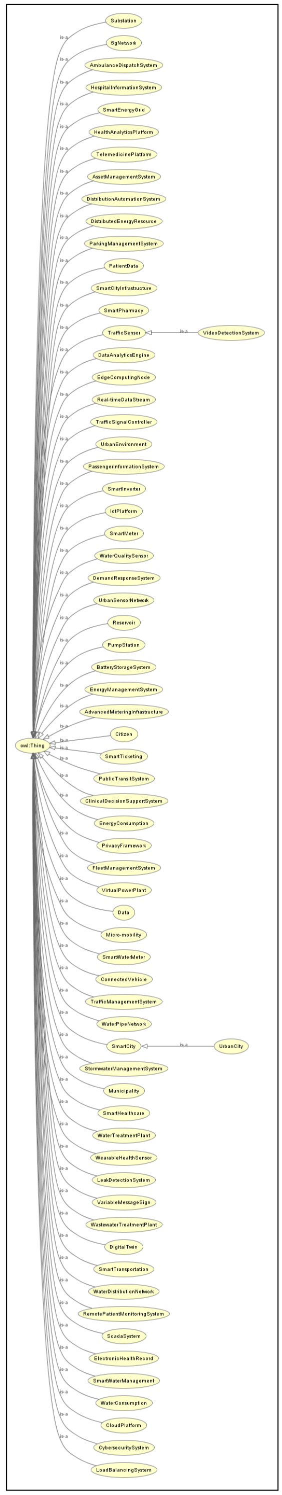


Figure 37: OWLViz visualisation of the complete Smart City Ontology class (Inferred) hierarchy, showing all 68 classes and their hierarchical relationships.

Figure 37 presents the OWLViz visualisation of the complete class inferred hierarchy. This diagram demonstrates the four-domain structure of the ontology and the cross-domain relationships enabled by shared IoT infrastructure classes.

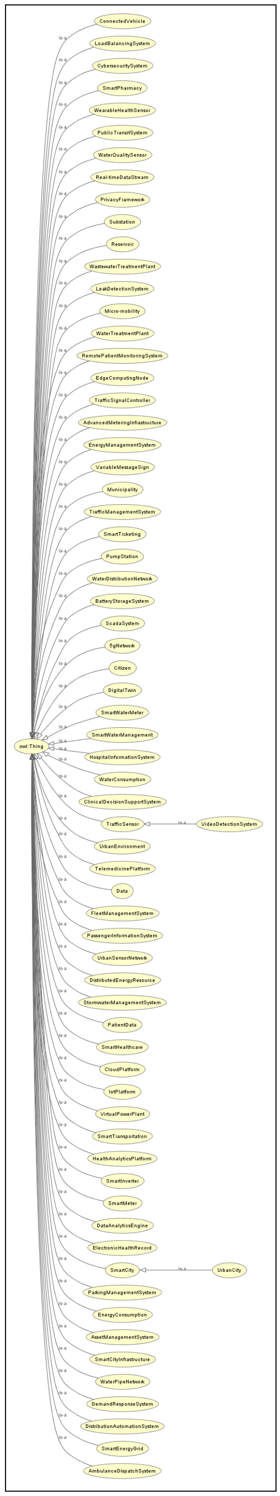


Figure 38: OWLViz visualisation of the complete Smart City Ontology class (Asserted) hierarchy, showing all 68 classes and their hierarchical relationships.

Figure 38 presents the OWLViz visualisation of the complete class asserted hierarchy. This diagram demonstrates the four-domain structure of the ontology and the cross-domain relationships enabled by shared IoT infrastructure classes.

5.3.4 Validation Summary

Table 2 summarises the validation of the 32 competency questions against the ontology.

Table 2: Competency Question Validation Summary.

CQ	Question	Answered By	Status
1	What is a Smart City?	SmartCity class + axiom 1	✓
2	What are the primary domains?	Axioms 2–5, isADomainOf	✓
3	What technologies enable a Smart City?	IoTPlatform, 5GNetwork, EdgeComputingNode	✓
4	What is Smart Transportation?	SmartTransportation class + axiom 2	✓
5	Components of Smart Transportation?	Axioms 22, 39, 48, 55	✓
6	How does traffic management work?	TrafficManagementSystem, axioms 23–33	✓
7	What smart mobility solutions exist?	Micro-mobility, ConnectedVehicle instances	✓
8	How is public transit integrated?	PublicTransitSystem, axioms 39–47	✓
9	How are electric vehicles supported?	electricVehicle, electricVehicleChargingStation	✓
10	What is a Smart Energy Grid?	SmartEnergyGrid class + axiom 3	✓
11	Components of a Smart Grid?	Axioms 59, 64, 68, 84	✓
12	How is energy consumption monitored?	SmartMeter, axiom 61	✓
13	How is renewable energy integrated?	DistributedEnergyResource, axioms 68–73	✓
14	How does demand response work?	DemandResponseSystem, axioms 75–76	✓
15	What is Smart Water Management?	SmartWaterManagement class + axiom 4	✓
16	Components of Smart Water system?	Axioms 93, 102, 111, 114, 117	✓
17	How is water quality monitored?	WaterQualitySensor, axiom 95	✓
18	How does leak detection work?	LeakDetectionSystem, axioms 108, 122	✓
19	How is wastewater treated?	WastewaterTreatmentPlant, axioms 114–116	✓
20	What is Smart Healthcare?	SmartHealthcare class + axiom 5	✓
21	Components of Smart Healthcare?	Axioms 130, 140, 143	✓
22	How are health records managed?	ElectronicHealthRecord, axioms 130–132	✓
23	What wearable devices are used?	WearableHealthSensor instances, axiom 134	✓
24	How does telemedicine function?	TelemedicinePlatform, axioms 132–133	✓
25	How are ambulances managed?	AmbulanceDispatchSystem, axioms 140–142	✓

Table 2 continued

CQ	Question	Answered By	Status
26	What is IoT in Smart City context?	IoTPlatform, axiom 6, transmitsDataVia	✓
27	What sensors are used?	TrafficSensor, WaterQualitySensor, WearableHealthSensor instances	✓
28	What is a Digital Twin?	DigitalTwin, axiom 10	✓
29	How is data collected and processed?	Axioms 6–9, UrbanSensorNetwork	✓
30	Privacy and security concerns?	CybersecuritySystem, PrivacyFramework, axioms 11, 13	✓
31	What are the benefits of a Smart City?	SmartCity, domain classes + optimises properties	✓
32	What are implementation challenges?	CybersecuritySystem, PrivacyFramework, integration axioms	✓

All 32 competency questions were successfully answered by the ontology through direct SPARQL querying, demonstrating that the scope and vocabulary were correctly defined.

6 Conclusion

Our study presents an OWL ontology covering the Smart City domain. We detailed the formalisation process, resulting in a structured knowledge base spanning four critical urban sectors: Smart Transportation, Smart Energy Grid, Smart Water Management, and Smart Healthcare, all underpinned by a shared IoT infrastructure backbone. The ontology consists of 68 classes, 38 object properties, and 85 named individuals, which are encoded in 186 formal axioms represented in both First-Order Logic and Description Logic.

We used the HermiT 1.4.3 reasoner in Protégé to confirm logical consistency across all classes and instances. Along with this, SPARQL queries successfully answered all 32 competency questions defined at the project's beginning, confirming that the ontology effectively supports knowledge retrieval and automated decision making.

This ontology provides a practical foundation for smart city information systems and real-world deployments. It serves as a shared vocabulary for developers building smart city knowledge graphs, interoperable IoT frameworks, and urban planning decision support systems.

Going forward, future work can expand the ontology to include things like social and economic features, citizen feedback systems, and real-time sensor integration through Sensor, Observation, Sample, and Actuator (SOSA) and Semantic Sensor Network (SSN) alignment.

Supporting Information

The following files are submitted alongside this report:

- **SmartCityOntology.owl**: The complete OWL ontology file in RDF/XML format, as developed in Protégé.

It should be noted that all relevant screenshots and competency question queries have been included within this document.

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